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L2 Metro Ring-Access With EVPN-ELAN

An HPE Juniper Networking Techpost

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HPE



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1 L2 Metro Ring-Access With EVPN-ELAN

Aris Georgakas - 10/29/2025

EVPN technology provides native CE direct L2 multi-homing capabilities, in either Active-Active or Active-Standby scheme. However, there might be a need for certain L2 access domains to form a ring topology around the EVPN PE endpoints. In our ACX7K JUNOS-EVO platforms, we leverage ERPS open-ring architecture (aka “non-vc-mode” G.8032v2 with subrings) to cover this use case and provide fast convergence under various failure conditions.

1.1 EVPN-ELAN ESI Multihoming Access Quick Refresh

EVPN is the protocol of choice for flexible and scalable L2 overlay connectivity for modern transport networks. A unique EVPN characteristic is that MAC address learning between PEs occurs in the BGP control plane. The PE routers exchange MAC and MAC-IP routes reachability information using multiprotocol-BGP and then encapsulate customer CEs’ L2 traffic for forwarding between PEs. Two main encapsulation schemes are supported: MPLS (RFC7432) and VXLAN (RFC8365).

One of the significant advantages of EVPN technology is multihoming capability. It enables you to connect a customer site to two or more PE devices, providing redundant connectivity, which means a CE device can be multihomed to different PE devices. EVPN treats these PE-CE connections within the same EVPN Instance (aka EVI) as belonging to the same Ethernet Segment. In other words, it assigns the same Ethernet Segment identifier (ESI) on the inter-connections towards the same L2 CE end-device.

In EVPN, using a standard Ethernet segment identifier (ESI) on CE attach-circuits terminating to both EVPN PEs, allows for establishing a loop-free topology towards the CE device and enables fast convergence. ESI is exchanged as EVPN route type-1, through BGP protocol established between PEs. Due to this common ESI attribute, the remote PE can detect an access failure and reroute the traffic towards its multihoming PE peer.

ESI LAG access can be configured as active-active (A-A) or active-standby (A-S), like MC-LAG predecessor multihoming access scheme.

The EVPN route type-4 is only advertised for CE multi-homing use case, i.e., when ESI has a non-zero value. It determines which EVPN PE terminating the ESI will become the Designated Forwarder (DF). DF is the only forwarder of BUM traffic in a particular Ethernet segment, and non-DF PEs block BUM traffic. The Type-4 route is advertised only between the multi-homing PE peers, terminating the Ethernet segment.

Note that the configured ESI value must be unique across all IFDs (i.e., ESI per major interface) and IFLs (i.e., ESI per sub-interface, if you wish to balance the service VLANs among the multihoming PEs). Features like EVPN “Egress-Link-Protection” and “Dynamic-List-Next-Hop” improve convergence times dramatically; however, it is not within the scope of this article to analyze them further.

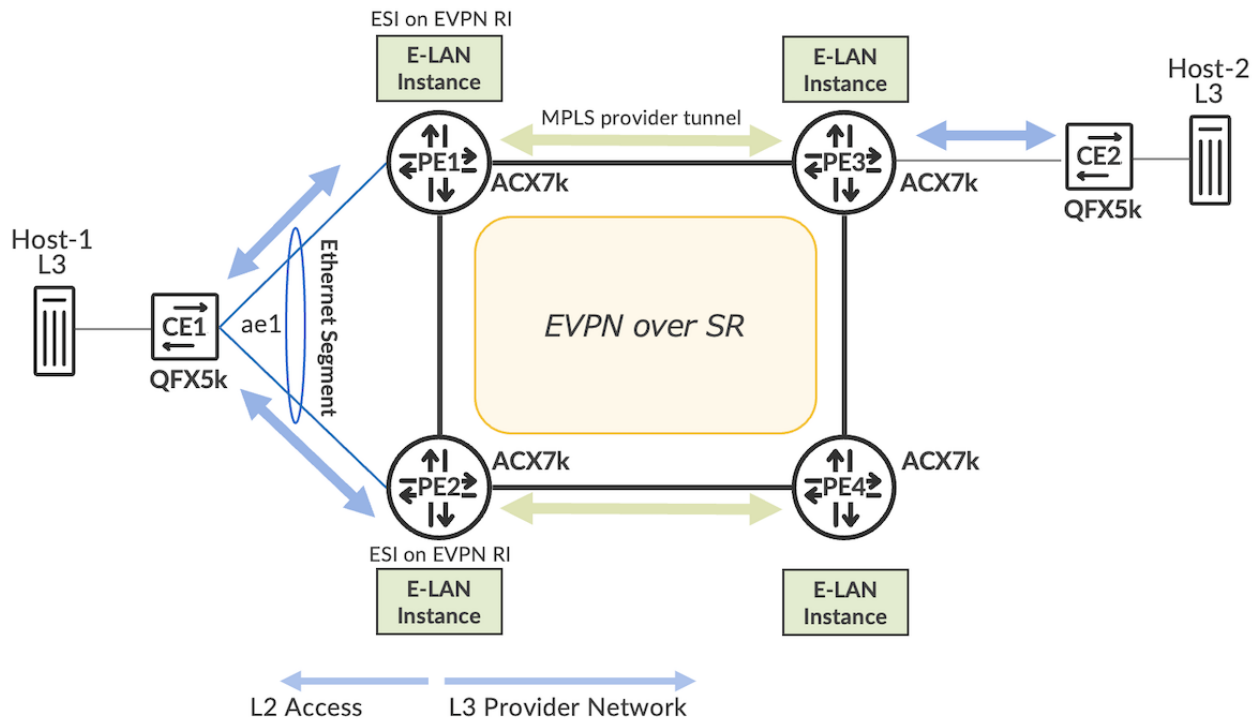


Figure 1: Typical EVPN Multi-Homing (MH) Direct ESI-LAG Access

Figure 1 illustrates a typical EVPN access ESI LAG scheme, where CE1 is EVPN multihomed (MH) to two PE routers with direct PE-CE connections. CE is unaware that it is connected to two individual PE devices and treats the multihoming connection as a LAG interface with LACP protocol activated optimally (optionally).

With respect to EVPN routing, CE2 has two overlay routing paths to reach CE1. The ESI Multihoming mode (A-A vs A-S) determines which PE router or both PE routers forward(s) traffic to the CE1. The PE router forwarding traffic to the CE1 device is termed Designated Forwarder (DF). If a failure occurs over this path, a new designated forwarder is elected to forward the traffic to CE1. The DF PE is responsible for forwarding BUM traffic towards multi-homed CE. When an ingress PE floods BUM traffic, it adds ESI label and prevents flood traffic from echoing back to the CE multi-homed Ethernet Segment (split-horizon filtering). ESI label is used only for BUM traffic.

Therefore, EVPN ESI multihoming helps to maintain EVPN service and traffic forwarding to and from the multihomed site in the event of the following types of network failures:

- PE to CE link failure
- PE node failure
- MPLS transport failure between a local PE and a remote PE

The typical EVPN-ELAN ESI MH access scenario described is familiar and common for most of us. This was a true evolution of the traditional MCLAG-to-VPLS ELAN decoupled mode, embedding multi-homing capability within the EVPN-ELAN protocol natively. Note also that BGP-VPLS has also native CE multi-homing capability (supported for both FEC 128 and FEC 129 auto-discovery flavors), however only active-standby MH access was feasible.

Nevertheless, in metro environments, we may still encounter L2 ring access topology, where multiple daisy-chained CE devices are East-West (E-W) connected towards the PE EVPN routers. xDWDM with e.g. Passive

MUX/DEMUX filters and colored optics or Add-Drop MUXes, could provide a solution in this case and emulate from each respective CE direct MH connections towards PEs, through common fiber Rx-Tx fiber infrastructure.

Still, though, there might be use cases where xDWDM workarounds cannot be implemented, due to additional cost, space/power constraints, etc., and another L2 loop-free fast-convergence solution needs to be deployed. Hence, stay tuned for the following sections!

1.2 L2 Ring-Access with PWE Overlays Challenges

L2 ring access domains to MPLS Core, is not a new topic of concern. With VPLS ELAN predecessor access it was always a metro design consideration. Original approach was to deploy xSTP protocol within the CEs ring topology and PEs participating on it, i.e. PEs being part and hold state of STP state machine. Then using PWEs to remote PEs locations, they could create overlay interconnects from one L2 access domain to another remote one. Unfortunately, this approach was always hiding loop risks for our Core network, since L2 control traffic is not forwarded by default over PWE overlays, meaning xSTP is not eligible to detect and block an end-to-end loop.

There are still ways to improve and fine-tune, obviously: e.g., allowing BPDU (tagged or untagged) control packets through dedicated PWEs overlays (L2PT activation may be required), or trying to interconnect MH PEs with a dedicated L2 link, etc. Again, these are not always feasible: e.g., my PEs are deployed within cabinets and physical inter-connections are not available; or very costly to implement, and still “hide” loop risks. This is especially true for L2 BUM traffic, which can accidentally be forwarded towards CE ring access domains, from both MH PE sides (due to L2 split-horizon flood rules), and then facing a catastrophic L2 loop condition.

To be fair, other vendors have developed some evolved technologies using PE L2 Access-Gateway technology (termed as AGs, e.g., MST-AG). This is a form of minimal xSTP implementation on the PEs’ side, where the PEs themselves do not really hold any xSTP state and do not need to handle any other BPDU - except TCNs. L2 access ring CEs still run xSTP, and through the AG PEs feature, we get a L2 loop-free e2e operation and optimal convergence. For various reasons, we never invested in JUNOS/EVO for this feature development.

One of them was that the EVPN-ELAN emerging protocol is becoming more and more popular, and it could solve most of these access design concerns natively. So let us focus on this!

1.3 L2 Ring-Access with ERPS-to-EVPN

What about modern EVPN and L2 ring access? Is there a simple way to have an L2 Ring-Access and be able to obtain e2e loop-free, scalable, and fast-convergence topologies over my MPLS transport? Can I still have the luxury of Anycast/VGA IRB floating Gateways, like in direct CE MH connectivity?

In the diagram below, we have an L2 ring access topology with three CE1, CE2, and CE3 L2 devices being daisy-chained, and the ring endpoints are East-West (E-W), terminating on PE1 and PE2 EVPN routers, respectively. In our physical topology, there is a physical back-to-back (b2b) link between PE1 and PE2. However, even if a b2b link is not feasible in a real-life metro aggregation topology, we could only rely on MPLS underlay connectivity through the upstream P/PE routers. The b2b local link usually facilitates lower latency, faster convergence times, and eliminates traffic “tromboning” upon node/link local failures. Just be aware that this

b2b link between PE1 and PE2 is a pure L3 MPLS link, i.e., it is not a L2 direct link for closing the L2 access ring.

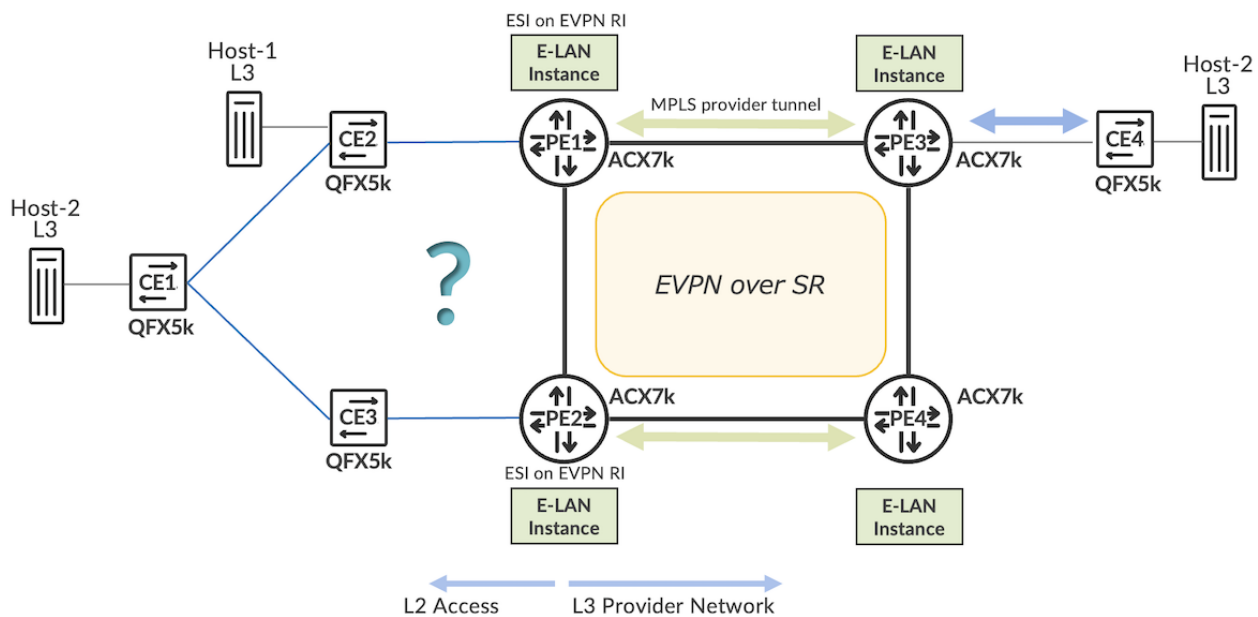


Figure 2: L2 Ring Topology EVPN E-W Access

Concern 1: Can I have EVPN ESI-LAG MH A-A balancing with L2 ring access? The simple answer is “No”. Main Reasons:

- For downstream traffic sourced by Host-3 towards the remote L2 access ring, PE3 will perform EVPN aliasing and load balance overlay L2 traffic across both PE1 and PE2. At the same time our L2 ring access may run a L2 protocol, like xSTP or ERPS, to eliminate L2 loops within the access domain and thus have a particular port blocked for this purpose. This may lead to partial L2 traffic blackholing if traffic destined to a specific CE enters the ring from the wrong PE side.
- L2 BUM downstream traffic towards the ring is only forwarded through the elected DF. But again, due to PE downstream traffic aliasing and any blocked ports within the ring, BUM traffic may never manage to reach certain CE1-CE2-CE3 nodes.
- Due to common ESI label filtering rules, L2 protocol TCN packets may not be able to be propagated through all the CE members of the ring,

Concern 2: Can I have EVPN ESI-LAG MH A-S with L2 ring access? This can only be used if the L2 access domain does not need to run any L2 protocol loop protection mechanism, and other methods are used to determine E-W L2 traffic flow from/to each CE under normal operation. Even under this condition - i.e. when there is no mechanism to determine E-W and we only rely on EVPN A-S BGP between PEs to eliminate the loop - this may lead to some suboptimal behavior during failovers and slow convergence. This is because Core PEs are totally “decoupled” and “agnostic” of the subtended access ring topology and forwarding state, not aware at all.

On the other hand, if we decide to run an L2 protocol on the L2 access ring (e.g. because we have a ladder ring access topology), then ESI-LAG MH A-S is not feasible at all. Main reasons in this case:

- For downstream traffic sourced by CE3 towards the remote L2 access ring, PE3 will steer overlay L2 traffic across towards the DF PE node (Modulo based or preference-based algorithms for DF selection); assuming PE1 for our Figure-2 diagram. At the same time, the L2 protocol, like xSTP or ERPS, will have a

particular port blocked to ensure L2 loop-free topology. This may lead to certain L2 traffic blackholing for one specific destination CE on the ring, since the blocked port in access will prohibit it from arriving at it. This applies to both unicast and BUM traffic flows.

- Due to common ESI label filtering rules, L2 protocol TCN packets may not be able to be propagated through all the CE members of the ring,

In fact, there is an IETF draft for the EVPN Ring access use-case, but it is not widely adopted at the moment. We do not support it on any JUNOS platform currently: [draft-brissette-bess-evpn-l2gw-proto-06](#)

So, what is happening? Is this a dead-end for L2 ring access with EVPN in JUNOS? Not exactly, since there is an available solution in JUNOS/EVO indeed, which can be implemented currently. **This is deploying ERPS Open-Ring (aka “non-vc-mode” G8032v2 with subrings) protocol for the L2 access ring terminating on EVPN PEs. The solution is qualified on ACX7000s and qualified with 24.4R1-EVO+.** Let us deep-dive further on this!

1.4 Why ERPS G8032v2 on L2 Access Ring?

In a nutshell, ERPS is an L2 protocol ensuring an L2 loop-free topology. There is a special link, called Ring Protection Link (RPL), that blocks and protects the ring topology, forming a L2 loop. That means that in regular operation, if there is no failure condition on the L2 ring, the RPL is in blocked state and does not pass any traffic. This RPL port is controlled by a ring node called the RPL owner.

There can be only one RPL owner per L2 ring access topology, and it is responsible for setting the state of the RPL port. This means that under failure conditions, the RPL owner will be responsible for unblocking it and start passing traffic. This is based on Automatic Protection Switching (APS) mechanism, which coordinates link failures and RPL block/unblock. When ring failure condition is recovered, then RPL owner sets RPL link back to block state, termed as revertive protection switching action. ERPS ring can be operated in revertive and non-revertive modes:

- Revertive Mode: traffic channel is restored to working path and RPL will be blocked, after The Wait to Block (WTB) timer expiry - under failure condition (default mode). We will also use this default revertive mode in our configuration setup.
- Non-Revertive mode: traffic channel continues to use RPL, until a clear command is issued. Can be enabled through configuration.

Therefore, ERPS caters within a L2 ring for loop avoidance and the utilization of learning, forwarding, and filtering database (FDB) mechanisms for switching traffic to an alternate path, upon failure/recovery conditions.

The **L2 nodes forming the ring topology have three different ERPS types:**

- **Normal node:** The node has no special role on the ring.
- **RPL owner node:** The node owns the RPL and blocks or unblocks traffic over the RPL.
- **RPL neighbor node:** The node that directly connects the RPL link of the RPL owner.

In addition, the **ERPS ring states** are:

- **init:** Not a participant of a specific ring.

- **idle:** No failure on the ring; the node is performing normally. For a normal node, traffic is unblocked on all ordinary ring ports. For the RPL owner or RPL neighbor, traffic is blocked on the ring port that connects to the RPL and unblocked on the other ring port.
- **protection:** A failure occurred on the ring. For a normal node, traffic is blocked on the ring port that connects to the failing link and unblocked on working ring ports. For the RPL owner, traffic is unblocked on both ring ports if they connect to non-failure links.
- **pending:** The node is recovering from failure or its state after a clear command is used to remove the previous manual command. When a protection group is configured, the node enters the pending state. When a node is in pending state, the WTR or WTB timer will be running. All nodes are in pending state till WTR or WTB timer expiry.
- **force switch:** A force switch is issued. When a force switch is issued on a node in the ring all nodes in the ring will move into the force switch state.
- **manual switch:** A manual switch is issued. When a manual switch is issued on a node in the ring all nodes in the ring will move into the manual switch state. Manual switch has lower priority than the Force switch. Manual switch will be cleared on applying force switch.

And **Ring protection conditions** and **commands** are:

- **Signal fail (SF):** When an SF condition is detected on a ring link and it is determined to be a “stable” failure, Ethernet ring nodes adjacent to the failed ring link initiate the protection switching mechanism.
- **No request (NR):** The condition when no local protection switching requests are active.
- **Forced switch (FS):** This command blocks a ring port on which the command is issued.
- **Manual switch (MS):** In the absence of a failure or FS, this command blocks a ring port on which the command is issued.
- **Clear:** The Clear command, at the Ethernet ring node, is used for the following operations:
 - Clearing an active local administrative command (e.g., Forced switch or Manual switch).
 - Triggering reversion before the WTR or WTB timer expires in case of revertive operation.
 - Triggering reversion in case of non-revertive operation.

1.4.1 ERPS/G8032v2 operation in a single Ethernet ring (upon protection failure condition):

- Normal operation steady state without any SF in the ring. All nodes in idle state.
- If a Failure occurs (e.g. Link between R3 & R4 failed as shown in Figure-3) Ethernet ring nodes adjacent to failure will detect a local signal failure condition and after respecting the hold-off time, block the failed ring port and perform the FDB flush.
- Ethernet ring nodes that detect Single Failure condition (SF), will start sending R-APS (SF) messages periodically with the (node ID,BPR) pair on both ring ports, while the SF condition persists.
- All Ethernet ring nodes receiving an R-APS (SF) message perform an FDB flush. When the RPL owner node R1 and RPL neighbor node R2 receive an R-APS (SF) message, they each unblock their end of the RPL and perform the FDB flush.
- Stable SF condition, R-APS (SF) messages on the Ethernet ring. Further R-APS (SF) messages trigger no further action.

Recovery:

- If the failed ring is recovered (between R3 & R4), the adjacent nodes will detect clearing of SF condition.

They start the guard timer and initiate the periodical transmission of R-APS (NR) messages on both ring ports. Note that guard time prevents the reception of R-APS messages.

- When the ring nodes receive an R-APS (NR) message, the (node-ID, BPR) pair of a receiving ring port is deleted and the RPL owner node starts the WTR timer.
- When the guard timer expires on R3 and R4, they will start accepting the new R-APS messages that they receive.
- Once WTR timer is expired, the RPL owner node blocks its end of the RPL sends an R-APS (NR, RB) message with the (node ID, BPR) pair performs the FDB flush.
- When R3 & R4 receives an R-APS (NR, RB) message, they remove the block ring ports and stop sending R-APS (NR) messages. In addition to this, ring nodes perform the FDB flush when receiving an R-APS (NR, RB) message.

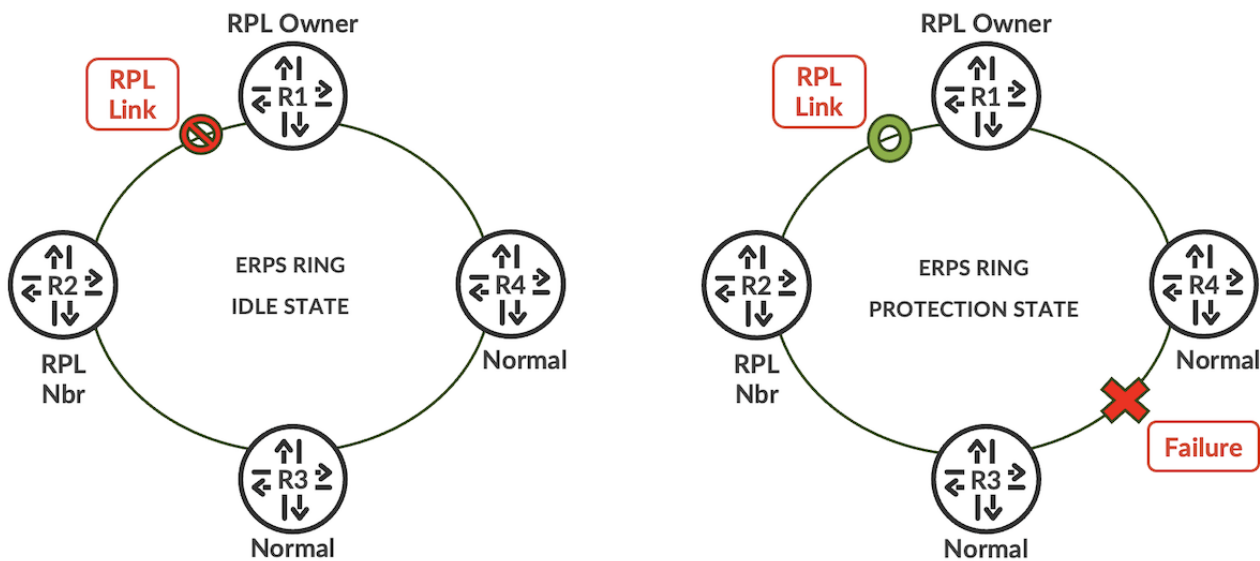


Figure 3: ERPS Normal Operation (Idle State) vs ERPS Signal Failure (Protection State) in Single-Ring Topology

Is that all we need to know for validation? In fact, we need one extra piece of info, and this is EPRS with primary and subtended secondary interconnected rings operation (aka G8032v2 operation). Typically, all ERPS nodes are connected as two closed rings connected (i.e., multi-ring topology), as depicted in the Figure 4 topology diagram shown below.

1.4.2 ERPS/G8032v2 operation in multi-Ethernet ring (upon protection failure condition):

All nodes are connected as two closed rings connected, termed as a ladder L2 ring network. The nodes common to two rings are called “Inter-connecting nodes”. The interconnecting link between these two nodes can be part of one instance. For example, here (depicted in the figure below), it is part of the major/primary ring.

Two rings can be interconnected in two different models:

- **With Virtual Channel:** Interconnected link is used for transmitting ERPS PDUs of sub ring.
- **Without Virtual Channel:** Interconnected link is not used for transmitting ERPS PDUs.

In JUNOS, we support only the Interconnection of rings without R-APS virtual channel. Each ring has its own RPL owner node and RPL that is blocked to avoid loop in the topology.

The G8032v2 operation for failure in major ring and sub ring is same as explained in a single ring scenario, i.e. failure in a ring blocks the failed ring port and sends the SF messages to all the nodes in the link. When RPL owner node receives these messages, it will flush and unblock the RPL end, and traffic will switch over.

Flush is pertained only to instance and any failure in sub ring will not impact or change direction in major ring. In some cases when the traffic is flowing across the rings, under a signal failure condition in subring, there are cases when traffic must change its direction even in major ring. This means the topology change notification has to be forwarded to major ring, which will trigger a flush in all nodes of major ring.

The following figures are representing **ERPS Normal Operation (Idle Sate) vs ERPS Signal Failure (Protection State) in Multi-Ring Topology**.

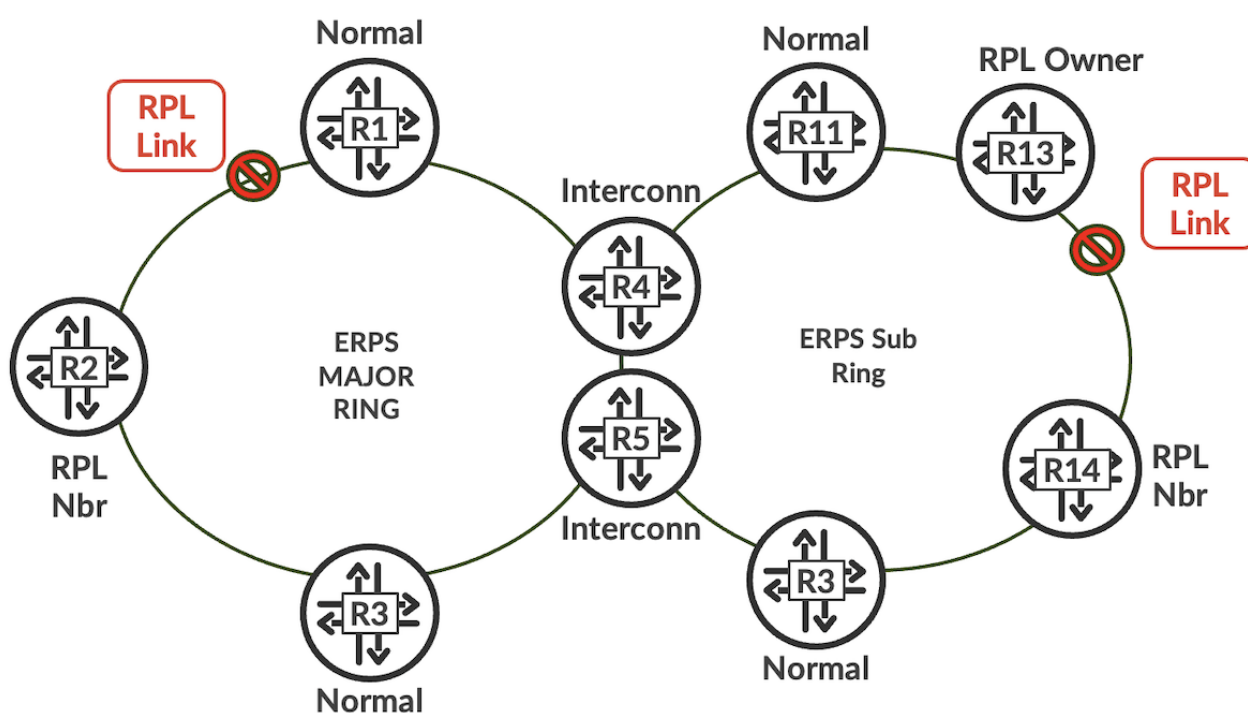


Figure 4: Idle state (normal operation)

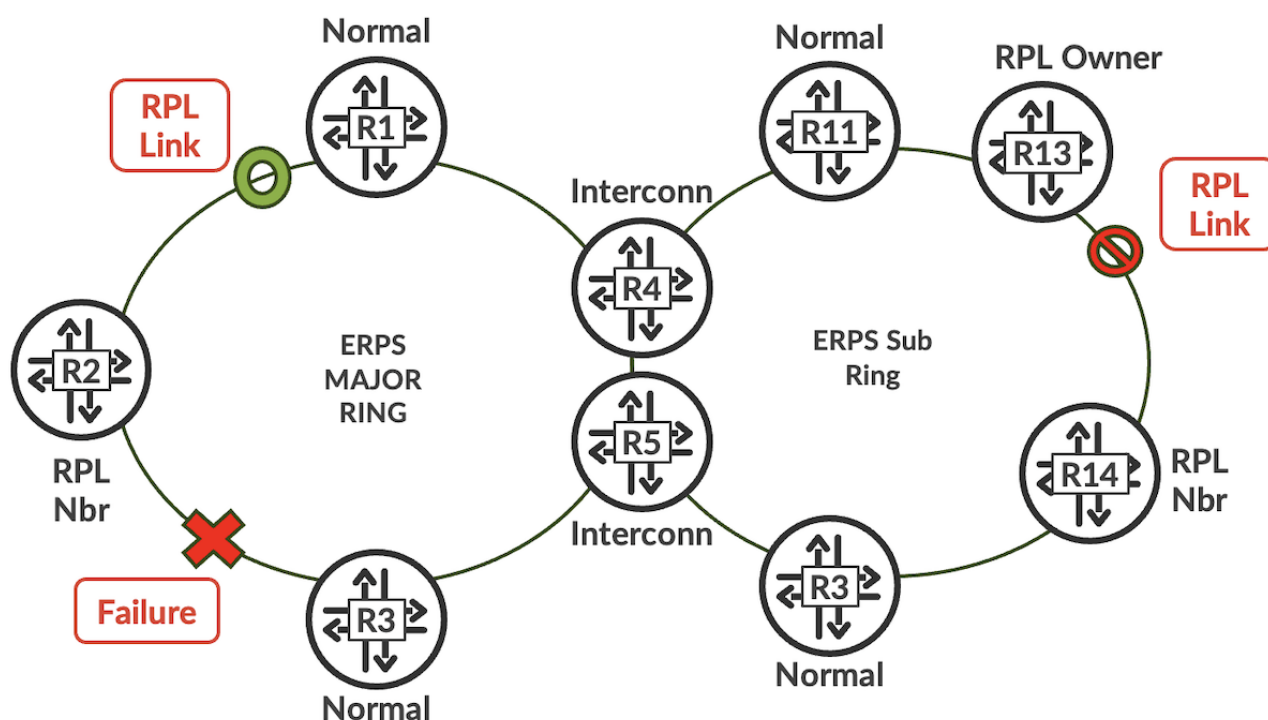


Figure 5: Failure condition in the Major Ring

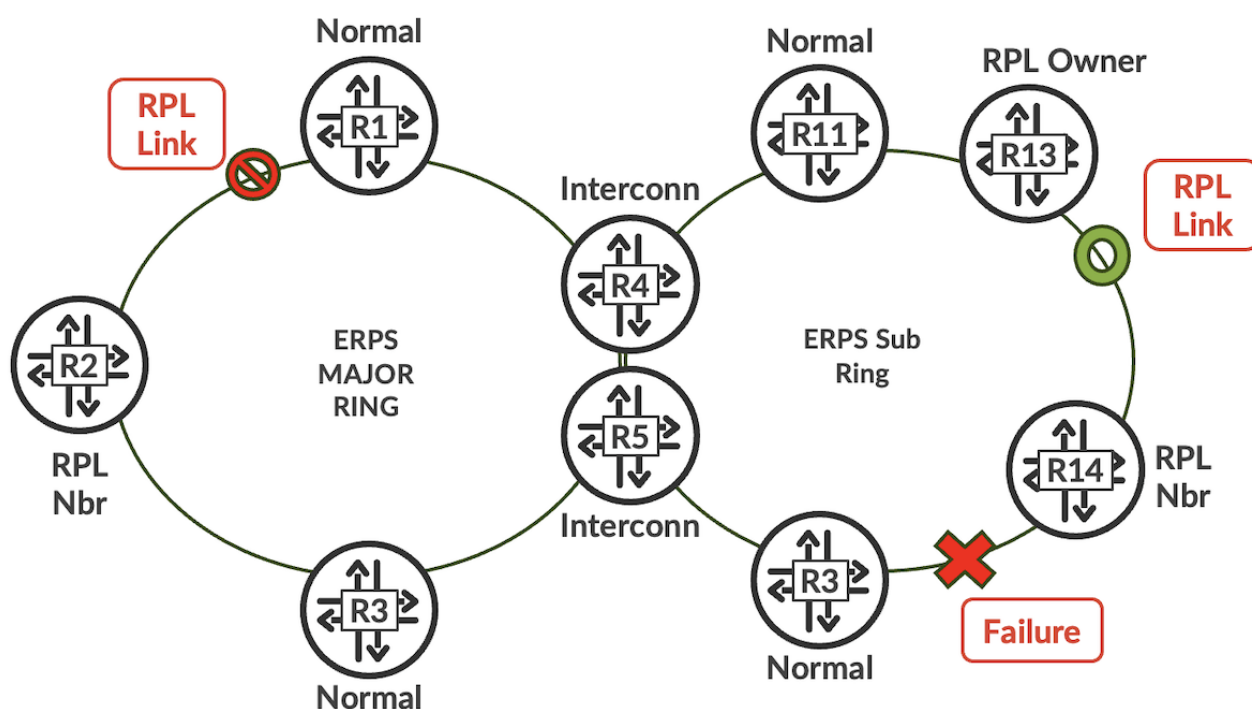


Figure 6: Failure condition in the Sub Ring

But why L2 Ethernet multi-rings are relevant here, we are dealing with EVPN ring access? In fact, EVPN ring access, leverages ERPS multi-ring topology behaviour! The L3 EVPN-MPLS domain is acting as the “Major” Ring, and the L2 CE ring access is acting like the “Sub” ring. More specific, the CE sub-ring is operating in non-virtual channel mode (**aka ethernet-ring non-vc-mode**), and all its L2 nodes are configured with this option, **aka deploying an ERPS open-ring alike architecture**.

EVPN PEs operate as “Interconnect routers” stitching ERPS L2 sub ring to the EVPN “major” ring domain, simulating an ERPS open-ring setup. Please note that both ACX7K PEs are learning the host end-points MAC addresses: one PE (e.g. PE2) learns it directly through L2 data plane mac learning; and the other PE (e.g. PE1) indirectly through PE2 EVPN bgp advertisement (since L2 path is RPL blocked). Therefore, in normal operation (ERPS idle state) traffic from host, flows through PE2 towards the EVPN MPLS core and vice-versa. When there is a failure condition within the CE L2 sub ring, and ERPS switches traffic to the opposite direction towards PE1 path, traffic convergence and packet loss are minimal. This is because PE1 is already aware of host MAC due to EVPN MAC syncing process and ERPS fast restoration actions (incl. mac-flush), within the ring access domain.

RPL owner unblocks the traffic over the RPL link. i.e. APS protocol blocks traffic on failed link and unblocks traffic on the RPL. When failure condition is cleared, revertive protection switching action blocks again the L2 traffic over the RPL and unblocks it towards the link which experienced the failure condition before.

1.5 ERPS-to-EVPN Normal Operation, Configuration & Validation

Let us verify and validate all the previous info in our simplified L2 ring access to EVPN-ELAN network topology.

We select CE2 (qfx5120-274):

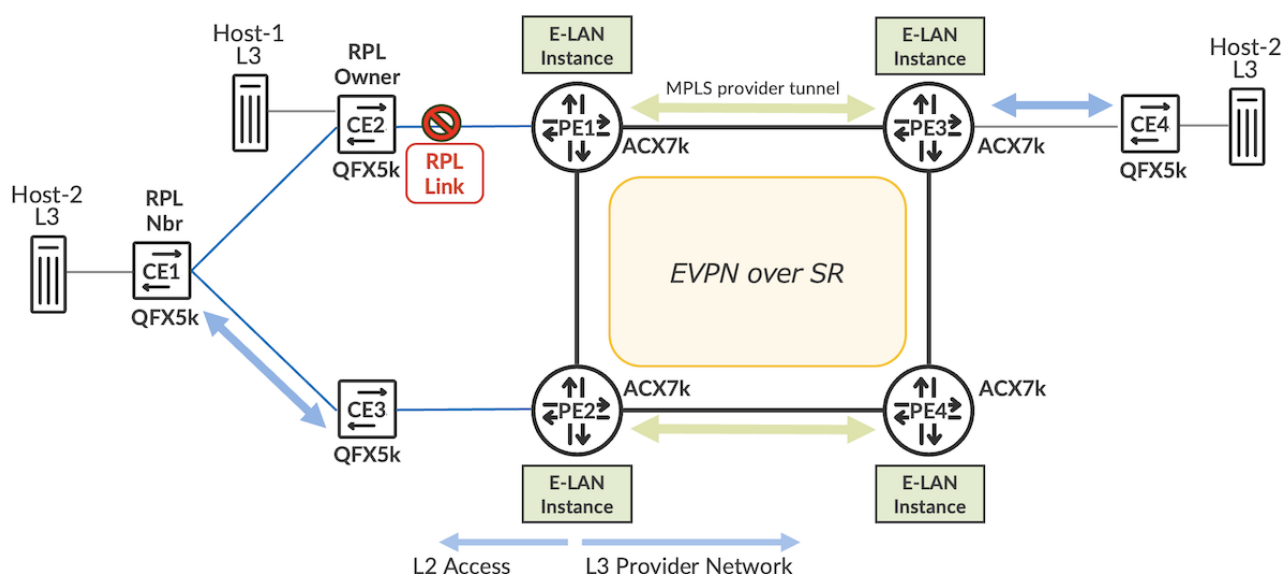


Figure 7: ERPS Normal Operation (Idle State), Validation Topology

In this topology QFX ring simulates the L2 access domain and we have only configured a single ERPS instance. With no failure on the ring under normal operation, ERPS instance has converged in IDLE state, after WTR expiration interval. RPL link, in between qfx5K-CE2 and acx7K-PE1 shall be in discard state.

Note: Administrators may decide to configure more ERPS instances, in case they desire to load balance their service vlans on E-W directions of the L2 access ring, towards both ACX7K PEs.

ERPS RPL-Owner CE2 Node (qfx5120-274)

```
1  ### L2 E-W Interfaces
2
```

```

3 interfaces {
4   et-0/0/48 {
5     flexible-vlan-tagging;
6     mtu 9000;
7     encapsulation flexible-ethernet-services;
8     unit 10 {
9       encapsulation vlan-bridge;
10      vlan-id 10;
11    }
12    unit 100 {
13      encapsulation vlan-bridge;
14      vlan-id 100;
15    }
16  }
17  et-0/0/49 {
18    flexible-vlan-tagging;
19    mtu 9000;
20    encapsulation flexible-ethernet-services;
21    unit 10 {
22      encapsulation vlan-bridge;
23      vlan-id 10;
24    }
25    unit 100 {
26      encapsulation vlan-bridge;
27      vlan-id 100;
28    }
29  }
30  ....
31 }
32
33 ### ERPS & VLANs
34
35 protocols {
36   lldp {
37     port-id-subtype interface-name;
38     neighbour-port-info-display port-id;
39     interface all;
40   }
41   protection-group {
42     ethernet-ring erp1 {
43       ring-protection-link-owner; ### QFX-CE2 is the RPL link-owner
44       ring-id 1;
45       non-vc-mode; ### Open-Ring architecture defined
46       east-interface {
47         control-channel {
48           et-0/0/49.10;
49         }
50       }
51       west-interface {
52         control-channel {
53           et-0/0/48.10;
54         }
55       }
56       ring-protection-link-end; ### RPL link towards ACX-PE1 direction
57       data-channel {
58         vlan 100;
59       }
60     }
61   }
62 }
63 vlans {
64   v10 {
65     interface et-0/0/48.10;
66     interface et-0/0/49.10;
67   }
68   v100 {
69     interface et-0/0/48.100;
70     interface et-0/0/49.100;
71     interface et-0/0/50.100;
72   }
73 }

```

ERPS Non-RPL CE3 Node (qfx5120-275) aka Normal Node

```

1 ### L2 E-W Interfaces
2
3 interfaces {
4   et-0/0/48 {
5     flexible-vlan-tagging;
6     mtu 9000;

```

```

7     encapsulation flexible-ethernet-services;
8     unit 10 {
9         encapsulation vlan-bridge;
10        vlan-id 10;
11    }
12    unit 100 {
13        encapsulation vlan-bridge;
14        vlan-id 100;
15    }
16 }
17 et-0/0/49 {
18     flexible-vlan-tagging;
19     mtu 9000;
20     encapsulation flexible-ethernet-services;
21     unit 10 {
22         encapsulation vlan-bridge;
23         vlan-id 10;
24     }
25     unit 100 {
26         encapsulation vlan-bridge;
27         vlan-id 100;
28     }
29 }
30 ....
31 }
32
33 ### ERPS & VLANs
34
35 protocols {
36     lldp {
37         port-id-subtype interface-name;
38         neighbour-port-info-display port-id;
39         interface all;
40     }
41     protection-group {
42         ethernet-ring erp1 {
43             ring-id 1;
44             non-vc-mode; ### Open-Ring architecture defined
45             east-interface {
46                 control-channel {
47                     et-0/0/48.10;
48                 }
49             }
50             west-interface {
51                 control-channel {
52                     et-0/0/49.10;
53                 }
54             }
55             data-channel {
56                 vlan 100;
57             }
58         }
59     }
60 }
61 vlans {
62     vl10 {
63         interface et-0/0/48.10;
64         interface et-0/0/49.10;
65     }
66     vl100 {
67         interface et-0/0/48.100;
68         interface et-0/0/49.100;
69         interface et-0/0/50.100;
70     }
71 }

```

1.5.1 Note: CE1 non-RPL node (qfx5120-268) has exactly similar configuration.

ACX7100-PE1 (acx7100-209)

```

1 interfaces {
2     ...
3     et-0/0/0 { ### MPLS Uplink towards Tail-End PE3
4         description to-acx7100-211;
5         mtu 9216;
6         unit 0 {

```

```

7      alias acx7100-211_et-0/0/0;
8      description "Connected to link acx7100-211:et-0/0/0.0-acx7100-209:et-0/0/0.0";
9      family inet {
10         address 10.100.0.61/31;
11     }
12     family inet6 {
13         address 2a0e:cb00:700b:100::61/127;
14     }
15     family mpls {
16         maximum-labels 8;
17     }
18 }
19 }
20 ...
21 et-0/0/7 {
22     description to-acx7100-210;
23     mtu 9216;
24     unit 0 {
25         description "Connected to link acx7100-210:et-0/0/7.0-acx7100-209:et-0/0/7.0";
26         family inet {
27             address 10.100.0.39/31;
28         }
29         family inet6 {
30             address 2a0e:cb00:700b:100::39/127;
31         }
32         family mpls {
33             maximum-labels 8;
34         }
35     }
36 }
37 et-0/0/10 {
38     flexible-vlan-tagging;
39     mtu 9000;
40     encapsulation flexible-ethernet-services;
41     unit 10 {
42         encapsulation vlan-bridge;
43         vlan-id 10;
44     }
45     unit 100 {
46         encapsulation vlan-bridge;
47         vlan-id 100;
48     }
49 }
50 ...
51 irb {
52     unit 100 {
53         virtual-gateway-accept-data;
54         proxy-arp;
55         family inet {
56             address 100.11.11.2/24 {
57                 preferred;
58                 virtual-gateway-address 100.11.11.1;
59             }
60         }
61         virtual-gateway-v4-mac 00:00:00:01:01:01;
62     }
63 }
64 lo0 {
65     unit 0 {
66         family inet {
67             address 145.1.1.5/32;
68         }
69         family inet6 {
70             address 2a0e:cb00:aaaa::5/128;
71         }
72     }
73 }
74 }
75 routing-instances {
76     evpnerps {
77         instance-type mac-vrf;
78         protocols {
79             evpn {
80                 encapsulation mpls;
81                 default-gateway no-gateway-community;
82             }
83         }
84         service-type vlan-based;
85         interface et-0/0/10.100;
86         route-distinguisher 145.1.1.5:11;
87         vrf-target target:100:11;

```



```

88     vlans {
89         vl100 {
90             vlan-id 100;
91             interface et-0/0/10.100;
92             l3-interface irb.100;
93         }
94     }
95 }
96 }
97 protocols {
98     bgp {
99         group to_rr {
100             local-address 145.1.1.5;
101             family inet {
102                 unicast;
103             }
104             family inet-vpn {
105                 unicast;
106             }
107             family inet6 {
108                 unicast;
109             }
110             family inet6-vpn {
111                 unicast;
112             }
113             family evpn {
114                 signaling;
115             }
116             peer-as 65100;
117             multipath;
118             bfd-liveness-detection {
119                 minimum-interval 300;
120                 multiplier 3;
121             }
122             neighbor 145.1.1.1; ##### RR-1
123             neighbor 145.1.1.2; ##### RR-2
124         }
125     }
126     mpls {
127         explicit-null;
128         label-range {
129             srgb-label-range 100000 109999;
130         }
131         ipv6-tunneling;
132         interface et-0/0/7.0;
133         interface et-0/0/0.0;
134     }
135     ospf {
136         backup-spf-options {
137             use-post-convergence-lfa;
138             use-source-packet-routing;
139         }
140         traffic-engineering {
141             l3-unicast-topology;
142             advertisement always;
143         }
144         source-packet-routing {
145             explicit-null;
146             node-segment ipv4-index 5;
147         }
148         area 0.0.0.0 {
149             interface lo0.0 {
150                 passive;
151                 bfd-liveness-detection {
152                     minimum-interval 300;
153                     multiplier 3;
154                     full-neighbors-only;
155                 }
156             }
157             interface et-0/0/7.0 {
158                 interface-type p2p;
159                 bfd-liveness-detection {
160                     minimum-interval 300;
161                     multiplier 3;
162                     full-neighbors-only;
163                 }
164                 post-convergence-lfa;
165             }
166             interface et-0/0/0.0 {
167                 interface-type p2p;
168                 bfd-liveness-detection {

```

```

169         minimum-interval 300;
170         multiplier 3;
171         full-neighbors-only;
172     }
173     post-convergence-lfa;
174     delay-measurement;
175 }
176 }
177 labeled-preference 6;
178 reference-bandwidth 1000g;
179 }
180 ...
181 l2-learning {
182     global-mac-table-aging-time 60;
183 }
184 lldp {
185     port-id-subtype interface-name;
186     neighbour-port-info-display port-id;
187     interface all;
188 }
189 protection-group {
190     ethernet-ring erp1 {
191         ring-id 1;
192         non-vc-mode; ### Open-Ring architecture defined
193         east-interface {
194             control-channel {
195                 et-0/0/10.10;
196             }
197         }
198         west-interface {
199             interface-none;
200         }
201         data-channel {
202             vlan 100;
203         }
204     }
205 }
206 }
207 vlans {
208     vl10 {
209         interface et-0/0/10.10;
210     }
211 }

```

Note: ACX7100 PE2 (acx7100-210) has exactly similar configuration.

Remote ACX7100-PE3 (acx7100-211)

```

1  interfaces {
2      ae0 { ### to ACX7100-PE4
3          mtu 9216;
4          aggregated-ether-options {
5              lacp {
6                  active;
7                  periodic fast;
8              }
9          }
10         unit 0 {
11             description "Connected to link acx7100-211:ae0.0-acx7100-212:ae0.0";
12             family inet {
13                 address 10.100.0.10/31;
14             }
15             family inet6 {
16                 address 2a0e:cb00:700b:100::10/127;
17             }
18             family mpls {
19                 maximum-labels 8;
20             }
21         }
22     }
23     ...
24     et-0/0/0 {
25         description to-acx7100-209;
26         mtu 9216;
27         unit 0 {
28             description "Connected to link acx7100-211:et-0/0/0.0-acx7100-209:et-0/0/0.0";
29             family inet {
30                 address 10.100.0.60/31;
31             }

```

```

32     family inet6 {
33         address 2a0e:cb00:700b:100::60/127;
34     }
35     family mpls {
36         maximum-labels 8;
37     }
38 }
39 }
40 et-0/0/12 { ### L2 link to Remote-Host-3
41     description qfx5120-272;
42     flexible-vlan-tagging;
43     mtu 9000;
44     encapsulation flexible-ethernet-services;
45     unit 100 {
46         encapsulation vlan-bridge;
47         vlan-id 100;
48     }
49 }
50 ...
51 irb {
52     unit 100 {
53         proxy-arp;
54         family inet {
55             address 100.11.11.100/24;
56         }
57     }
58 }
59 lo0 {
60     unit 0 {
61         family inet {
62             address 145.1.1.3/32;
63         }
64         family inet6 {
65             address 2a0e:cb00:aaaa::3/128;
66         }
67     }
68 }
69 }
70 routing-instances {
71     evpnerps {
72         instance-type mac-vrf;
73         protocols {
74             evpn {
75                 encapsulation mpls;
76             }
77         }
78         service-type vlan-based;
79         interface et-0/0/12.100;
80         route-distinguisher 145.1.1.3:11;
81         vrf-target target:100:11;
82         vlans {
83             vl100 {
84                 vlan-id 100;
85                 interface et-0/0/12.100;
86                 l3-interface irb.100;
87             }
88         }
89     }
90 }
91 protocols {
92     bgp {
93         group to_rr {
94             local-address 145.1.1.3;
95             family inet {
96                 unicast;
97             }
98             family inet-vpn {
99                 unicast;
100            }
101            family inet6 {
102                unicast;
103            }
104            family inet6-vpn {
105                unicast;
106            }
107            family evpn {
108                signaling;
109            }
110            peer-as 65100;
111            multipath;
112            bfd-liveness-detection {

```

```

113         minimum-interval 300;
114         multiplier 3;
115     }
116     neighbor 145.1.1.1;
117     neighbor 145.1.1.2;
118 }
119 }
120 mpls {
121     label-range {
122         srgb-label-range 100000 109999;
123     }
124     ipv6-tunneling;
125     interface ae0.0;
126     interface et-0/0/0.0;
127 }
128 ospf {
129     backup-spf-options {
130         use-post-convergence-lfa;
131         use-source-packet-routing;
132     }
133     traffic-engineering {
134         l3-unicast-topology;
135         advertisement always;
136     }
137     source-packet-routing {
138         explicit-null;
139         node-segment ipv4-index 3;
140     }
141     area 0.0.0.0 {
142         interface lo0.0 {
143             passive;
144             bfd-liveness-detection {
145                 minimum-interval 300;
146                 multiplier 3;
147                 full-neighbors-only;
148             }
149         }
150         interface et-0/0/0.0 {
151             interface-type p2p;
152             bfd-liveness-detection {
153                 minimum-interval 300;
154                 multiplier 3;
155                 full-neighbors-only;
156             }
157             post-convergence-lfa;
158             delay-measurement;
159         }
160         interface ae0.0 {
161             interface-type p2p;
162             bfd-liveness-detection {
163                 minimum-interval 300;
164                 multiplier 3;
165                 full-neighbors-only;
166             }
167             post-convergence-lfa;
168             delay-measurement;
169         }
170     }
171     labeled-preference 6;
172     reference-bandwidth 1000g;
173 }
174 l2-learning {
175     global-mac-table-aging-time 60;
176 }
177 lldp {
178     port-id-subtype interface-name;
179     neighbour-port-info-display port-id;
180     interface all;
181 }
182 }

```

Note: ACX7100-PE4 is only acting as transit LSR (P router) for this setup.

There are two CE hosts on the L2 access ring (Host-1,2) and a third remote end host on the other side of the EVPN-MPLS transport network:

- Host-1: IPv4 100.11.11.10; MAC 88:30:37:6e:a1:c3 (attached to CE2)

- Host-2: IPv4 100.11.11.20; MAC 88:30:37:6e:a1:db (attached to CE1)
- Host-3: IPv4 100.11.11.110; MAC e4:23:3c:9d:73:41 (attached to CE4)

Just for interconnectivity connectivity checking purposes, we have also defined on vlan-100 (Data VLAN), IRB interfaces within the ACX7K MAC-VRF EVIs:

- ACX7100-PE1 IRB.100: IPv4 100.11.11.2; MAC 68:22:8e:1d:9b:a6
- ACX7100-PE2 IRB.100: IPv4 100.11.11.3; MAC 68:22:8e:1d:ab:b6
- PE1&PE2 VGA-IRB: VIP 100.11.11.1; VMAC 00:00:00:01:01:01 (static)
- ACX7100-PE3 IRB.100: IPv4 100.11.11.100; MAC ac:78:d1:22:3b:82

1.5.2 RPL owner CE2 switch

```

1 aris@qfx5120-274# run show protection-group ethernet-ring node-state detail
2 Ethernet-Ring name : erp1
3 APS State : idle ### idle state in normal operation
4 Event : NR-RB
5 Ring Protection Link Owner : Yes
6 Wait to Restore Timer : disabled
7 Wait to Block Timer : disabled
8 Guard Timer : disabled
9 Operation state : operational
10
11 {master:0}[edit]
12 aris@qfx5120-274# run show protection-group ethernet-ring interface detail
13
14 Ethernet ring port parameters for protection group erp1
15
16 Interface name : et-0/0/49
17 Control channel name : et-0/0/49.10
18 Interface direction : east
19 Ring Protection Link End : No
20 Signal Failure : Clear
21 Forward State : forwarding ### forwarding normal operation mode
22 Interface Admin State : IFF ready
23
24 Interface name : et-0/0/48
25 Control channel name : et-0/0/48.10
26 Interface direction : west
27 Ring Protection Link End : Yes
28 Signal Failure : Clear
29 Forward State : discarding ### RPL port in discard state, to eliminate loop
30 Interface Admin State : IFF ready
31
32 {master:0}[edit]
33 aris@qfx5120-274# run show ethernet-switching table
34
35 MAC flags (S - static MAC, D - dynamic MAC, L - locally learned, P - Persistent static, C -
   Control MAC
36     SE - statistics enabled, NM - non configured MAC, R - remote PE MAC, O - ovsdb MAC
37     GBP - group based policy)
38
39 Ethernet switching table : 5 entries, 5 learned
40 Routing instance : default-switch
41   Vlan MAC MAC Age GBP Logical NH RTR
42   name address flags Tag interface Index ID
43   v100 00:00:00:01:01:01 D - et-0/0/49.100 0 0
44   v100 68:22:8e:1d:9b:a6 D - et-0/0/49.100 0 0
45   v100 68:22:8e:1d:ab:b6 D - et-0/0/49.100 0 0
46   v100 88:30:37:6e:a1:c3 D - et-0/0/50.100 0 0
47   v100 88:30:37:6e:a1:db D - et-0/0/49.100 0 0

```

Non-RPL owner CE3 switch

```

1 aris@qfx5120-275# run show protection-group ethernet-ring node-state detail
2 Ethernet-Ring name : erp1
3 APS State : idle
4 Event : NR-RB
5 Ring Protection Link Owner : No
6 Wait to Restore Timer : disabled
7 Wait to Block Timer : disabled

```

```

 8 Guard Timer : disabled
 9 Operation state : operational
10
11 {master:0}[edit]
12 aris@qfx5120-275# run show protection-group ethernet-ring interface detail
13
14 Ethernet ring port parameters for protection group erp1
15
16 Interface name : et-0/0/48
17 Control channel name : et-0/0/48.10
18 Interface direction : east
19 Ring Protection Link End : No
20 Signal Failure : Clear
21 Forward State : forwarding ### forwarding normal operation mode
22 Interface Admin State : IFF ready
23 Interface name : et-0/0/49
24 Control channel name : et-0/0/49.10
25 Interface direction : west
26 Ring Protection Link End : No
27 Signal Failure : Clear
28 Forward State : forwarding ### forwarding normal operation mode
29 Interface Admin State : IFF ready
30
31 {master:0}[edit]
32 aris@qfx5120-275# run show ethernet-switching table
33
34 MAC flags (S - static MAC, D - dynamic MAC, L - locally learned, P - Persistent static, C -
    Control MAC
35     SE - statistics enabled, NM - non configured MAC, R - remote PE MAC, O - ovsdb MAC
36     GBP - group based policy)
37
38 Ethernet switching table : 7 entries, 7 learned
39 Routing instance : default-switch
40   Vlan MAC MAC Age GBP Logical NH RTR
41   name address flags Tag interface Index ID
42   vl10 e4:23:3c:9d:91:d0 D - et-0/0/49.10 0 0
43   vl100 00:00:00:01:01:01 D - et-0/0/48.100 0 0
44   vl100 68:22:8e:1d:9b:a6 D - et-0/0/48.100 0 0
45   vl100 68:22:8e:1d:ab:b6 D - et-0/0/48.100 0 0
46   vl100 88:30:37:6e:a1:c3 D - et-0/0/49.100 0 0
47   vl100 88:30:37:6e:a1:db D - et-0/0/49.100 0 0
48   vl100 e4:23:3c:9d:73:41 D - et-0/0/48.100 0 0

```

Monitoring ERPS and MAC-VRF EVIs status of ACX7100 PEs under normal operation, is investigated. **Please note that the CE L2 ring facing interfaces are ESI single-homed, i.e. the L2 access ring is not treated as a common ESI multi-homing segment.** It worth also noting that both PE nodes from EPRS point-of-view are operating in non-virtual channel mode, aka open ring architecture.

All host MAC addresses are learned in ACX-PE1 (acx7100-209), through EVPN MAC/MAC-IP bgp learning process:

```

 1 aris@acx7100-209# show interfaces et-0/0/10 | display inheritance no-comments
 2 flexible-vlan-tagging;
 3 mtu 9000;
 4 encapsulation flexible-ethernet-services;
 5 unit 10 {
 6     encapsulation vlan-bridge;
 7     vlan-id 10;
 8 }
 9 unit 100 {
10     encapsulation vlan-bridge;
11     vlan-id 100;
12 }
13
14 ### PE1 ERPS State
15
16 [edit]
17 aris@acx7100-209# run show protection-group ethernet-ring node-state detail
18 Ethernet-Ring name : erp1
19 APS State : idle
20 Event : NR-RB
21 Ring Protection Link Owner : No
22 Wait to Restore Timer : disabled
23 Wait to Block Timer : disabled
24 Guard Timer : disabled
25 Operation state : operational

```

```

26
27 [edit]
28 aris@acx7100-209# run show protection-group ethernet-ring interface detail
29
30 Ethernet ring port parameters for protection group erp1
31
32 Interface name : et-0/0/10
33 Control channel name : et-0/0/10.10
34 Interface direction : east
35 Ring Protection Link End : No
36 Signal Failure : Clear
37 Forward State : forwarding ### Forwarding normal operation mode (Ring CE side)
38 Interface Admin State : IFF ready
39
40 Interface name : Unused ### Non-vc-mode open ring operation (MPLS Core)
41 Control channel name : Unused
42 Interface direction : west
43 Ring Protection Link End : NA
44 Signal Failure : NA
45 Forward State : NA
46 Interface Admin State : NA
47
48 ### PE1 EVPN State
49
50 [edit]
51 aris@acx7100-209# run show evpn instance extensive evpnerps
52 Instance: evpnerps
53   Route Distinguisher: 145.1.1.5:11
54   VLAN ID: 100
55   Per-instance MAC route label: 16
56   Control word enabled
57   Duplicate MAC detection threshold: 5
58   Duplicate MAC detection window: 180
59   MAC database status Local Remote
60     MAC advertisements: 1 6
61     MAC+IP advertisements: 2 6
62     Default gateway MAC advertisements: 2 1
63   Number of local interfaces: 2 (2 up)
64     Interface name ESI Mode Status AC-Role
65     .local..53 00:00:00:00:00:00:00:00:00:00 single-homed Up Root
66     et-0/0/10.100 00:00:00:00:00:00:00:00:00:00 single-homed Up Root
67   Number of IRB interfaces: 1 (1 up)
68     Interface name VLAN VNI Status L3 context
69     irb.100 100 Up master
70   Number of protect interfaces: 0
71   Number of bridge domains: 1
72     VLAN Domain-ID Intfs/up IRB-intf Mode MAC-sync v4-SG-sync v6-SG-sync
73     100 1 1 irb.100 Extended Enabled Disabled Disabled
74   Number of neighbors: 2
75     Address MAC MAC+IP AD IM ES Leaf-label DCI-Peer Flow-label DT2U-SID DT2M-SID
76     145.1.1.3 2 2 0 1 0 NO
77     145.1.1.6 4 4 1 1 0 NO
78   Number of ethernet segments: 1
79     ESI: 05:00:00:fe:4c:00:00:00:64:00
80     State-Bitfield: 0x43
81     ESI Route Label: 18
82     ESI Refcount: 1
83     ESI Num Macs: 1, ESI Num SGDBs: 0
84     Token-Route NH: 0
85     Number of Local interfaces: 1
86     Local interface: irb.100, Status: Up/Forwarding
87     Number of remote PEs connected: 1
88     Remote-PE MAC-label Aliasing-label Mode
89     145.1.1.6 17 0 all-active
90     DF Election Algorithm: MOD based
91     Designated forwarder: DF not elected yet
92     Last designated forwarder update: Never
93     Advertised split horizon label: 9424
94   SMET Forwarding: Disabled
95   RIB Table-ID: 184549386, Kernel Table-ID: 53, Kernel Table-Generation: 0
96   EVPN instance flags: 0x801810000
97   RTT Update Timestamp: Jan 17 12:33:23.160 2025
98   L2ALD state change Timestamp: Jan 17 12:33:23.817 2025
99   Core-Isolation change TS: Apr 8 15:13:03.792 2025, Core-Isolated: N
100   Last Core-Isolation Change Reason: evpn-peer-transition
101
102 [edit]
103 aris@acx7100-209#
104
105 ### All Host MAC addresses are learned through EVPN MAC syncing, since CE2 direct ring uplink is
    blocked by ERPS RPL.

```

```

106 [edit]
107 aris@acx7100-209# run show mac-vrf forwarding mac-table instance evpnerps
108
109 MAC flags (S - static MAC, D - dynamic MAC, L - locally learned, P - Persistent static, C -
110 Control MAC
111 SE - statistics enabled, NM - non configured MAC, R - remote PE MAC, O - ovsdb MAC
112 GBP - group based policy, B - Blocked MAC)
113
114 Ethernet switching table : 6 entries, 6 learned
115 Routing instance : evpnerps
116 Vlan MAC MAC Age GBP Logical NH MAC RTR
117 name address flags Tag interface Index property ID
118 vl100 00:00:00:01:01:01 DCP - 8421 8421
119 vl100 68:22:8e:1d:ab:b6 DCP - 8342 8342
120 vl100 88:30:37:6e:a1:c3 DC - 8342 8342 ### Host-1 MAC
121 vl100 88:30:37:6e:a1:db DC - 8342 8342 ### Host-2 MAC
122 vl100 ac:78:d1:22:3b:82 DC - 8083 8083
123 vl100 e4:23:3c:9d:73:41 DC - 8083 8083 ### Host-3 MAC
124
125 ### MAC-IP routes
126
127 [edit]
128 aris@acx7100-209# run show mac-vrf forwarding mac-ip-table instance evpnerps
129
130 MAC IP flags (S - Static, D - Dynamic, L - Local , R - Remote, Lp - Local Proxy,
131 Rp - Remote Proxy, K - Kernel, RT - Dest Route, (N)AD - (Not) Advt to remote,
132 RE - Re-ARP/ND, RO - Router, OV - Override, Ur - Unresolved, B - Blocked,
133 RTS - Dest Route Skipped, RGW - Remote Gateway, GBP - Group Based Policy,
134 RTF - Dest Route Forced, SC - Static Config, P - Probe, NLC - No Local Config,
135 LD - Local Down)
136 Routing instance : evpnerps
137 Bridging domain : vl100
138 IP MAC Flags GBP Logical Active
139 address address Tag Interface source
140 100.11.11.1 00:00:00:01:01:01 S,K irb.100 05:00:00:fe:4c:00:00:00:64:00
141 100.11.11.2 68:22:8e:1d:9b:a6 S,K irb.100
142 100.11.11.3 68:22:8e:1d:ab:b6 SR,K,RT 145.1.1.6
143 100.11.11.10 88:30:37:6e:a1:c3 DR,K,RT 145.1.1.6
144 100.11.11.20 88:30:37:6e:a1:db DR,K,RT 145.1.1.6
145 100.11.11.100 ac:78:d1:22:3b:82 SR,K,RT,RGW 145.1.1.3
146 100.11.11.110 e4:23:3c:9d:73:41 DR,K,RT 145.1.1.3
147
148 [edit]
149 aris@acx7100-209# run show route forwarding-table table evpnerps | find 88:30:37:6e:a1:c3 ###
150 LOCAL HOST-1
151 88:30:37:6e:a1:c3/48 user 0 indr 8342 1
152 ulst 8920 1
153 sftw Push 17 8890 1 et-0/0/7.0
154 10.100.0.38 ucst 1001 1 et-0/0/7.0
155 sftw Push 17, Push 100006(top) 8914 1 et-0/0/0.0
156 10.100.0.60 ucst 1000 1 et-0/0/0.0
157 <output ommited...>
158
159 [edit]
160 aris@acx7100-209# run show route table mpls.0 label 100006 ### explicit-null is configured for
161 SR-MPLS
162
163 mpls.0: 45 destinations, 46 routes (45 active, 0 holddown, 0 hidden)
164 + = Active Route, - = Last Active, * = Both
165
166 100006 *[L-OSPF/6/5] 00:04:32, metric 10
167 > to 10.100.0.38 via et-0/0/7.0, Swap 0
168 to 10.100.0.60 via et-0/0/0.0, Swap 100006
169 100006(S=0) *[L-OSPF/6/5] 00:04:32, metric 10
170 > to 10.100.0.38 via et-0/0/7.0, Pop
171 to 10.100.0.60 via et-0/0/0.0, Swap 100006
172
173 [edit]
174 aris@acx7100-209# run show route table mpls.0 label 17
175
176 mpls.0: 45 destinations, 46 routes (45 active, 0 holddown, 0 hidden)
177 + = Active Route, - = Last Active, * = Both
178
179 17 *[EVPN/7] 37w2d 06:16:54, routing-instance evpnerps, route-type Ingress-MAC, vlan-id 100
180 to table evpnerps.evpn-mac.0
181 [EVPN/7] 37w2d 06:16:54, routing-instance evpnerps, route-type Ingress-MAC, vlan-
182 id 100
183 to table evpnerps.evpn-mac.0
184
185 ### Similar outputs for LOCAL HOST2 MAC: 88:30:37:6e:a1:db (omitted for brevity)#####

```



```

183 [edit]
184 aris@acx7100-209# run show route forwarding-table table evpnerps | find e4:23:3c:9d:73:41 ###
      REMOTE HOST-3
185 e4:23:3c:9d:73:41/48 user 0 indr 8083 1
186                               ulst 8937 1
187                               sftw Push 17 8888 1 et-0/0/0.0
188                               10.100.0.60 ucst 1000 1 et-0/0/0.0
189                               sftw Push 17, Push 100003(top) 8932 1 et-0/0/7.0
190                               10.100.0.38 ucst 1001 1 et-0/0/7.0
191
192 <output ommited...>
193 [edit]
194 aris@acx7100-209# run show route table mpls.0 label 100003 ### explicit-null is configured for
      SR-MPLS
195
196 mpls.0: 45 destinations, 46 routes (45 active, 0 holddown, 0 hidden)
197 + = Active Route, - = Last Active, * = Both
198
199 100003 *[L-OSPF/6/5] 00:41:53, metric 10
200         > to 10.100.0.60 via et-0/0/0.0, Swap 0
201         to 10.100.0.38 via et-0/0/7.0, Swap 100003
202 100003(S=0) *[L-OSPF/6/5] 00:41:53, metric 10
203         > to 10.100.0.60 via et-0/0/0.0, Pop
204         to 10.100.0.38 via et-0/0/7.0, Swap 100003
205
206 [edit]
207 aris@acx7100-209# run show route table mpls.0 label 17
208
209 mpls.0: 45 destinations, 46 routes (45 active, 0 holddown, 0 hidden)
210 + = Active Route, - = Last Active, * = Both
211
212 17 *[EVPN/7] 37w2d 06:52:08, routing-instance evpnerps, route-type Ingress-MAC, vlan-id 100
213     to table evpnerps.evpn-mac.0
214     [EVPN/7] 37w2d 06:52:08, routing-instance evpnerps, route-type Ingress-MAC, vlan-
        id 100
        to table evpnerps.evpn-mac.0
215
216 [edit]
217 aris@acx7100-209# run show route advertising-protocol bgp 145.1.1.1 table evpnerps.evpn.0
218
219 evpnerps.evpn.0: 20 destinations, 35 routes (20 active, 0 holddown, 0 hidden)
220 Prefix Nexthop MED Lclpref AS path
221 2:145.1.1.5:11::0::00:00:00:01:01:01/304 MAC/IP
222 * Self 100 I
223 2:145.1.1.5:11::0::68:22:8e:1d:9b:a6/304 MAC/IP
224 * Self 100 I
225 2:145.1.1.5:11::0::00:00:00:01:01:01::100.11.11.1/304 MAC/IP ### VGA-VMAC
226 * Self 100 I
227 2:145.1.1.5:11::0::68:22:8e:1d:9b:a6::100.11.11.2/304 MAC/IP ### IRB ACX-PE1 System MAC
228 * Self 100 I
229 3:145.1.1.5:11::0::145.1.1.5/248 IM
230 * Self 100 I
231
232 [edit]
233 aris@acx7100-209# run show route receive-protocol bgp 145.1.1.1 table evpnerps.evpn.0
234
235 evpnerps.evpn.0: 20 destinations, 35 routes (20 active, 0 holddown, 0 hidden)
236 Prefix Nexthop MED Lclpref AS path
237 1:145.1.1.6:0::050000fe4c0000006400::FFFF:FFFF/192 AD/ESI
238 * 145.1.1.6 100 I
239 2:145.1.1.3:11::0::ac:78:d1:22:3b:82/304 MAC/IP
240 * 145.1.1.3 100 I
241 2:145.1.1.3:11::0::e4:23:3c:9d:73:41/304 MAC/IP
242 * 145.1.1.3 100 I
243 2:145.1.1.6:11::0::00:00:00:01:01:01/304 MAC/IP
244 * 145.1.1.6 100 I
245 2:145.1.1.6:11::0::68:22:8e:1d:ab:b6/304 MAC/IP
246 * 145.1.1.6 100 I
247 2:145.1.1.6:11::0::88:30:37:6e:a1:c3/304 MAC/IP
248 * 145.1.1.6 100 I
249 2:145.1.1.6:11::0::88:30:37:6e:a1:db/304 MAC/IP
250 * 145.1.1.6 100 I
251 2:145.1.1.3:11::0::ac:78:d1:22:3b:82::100.11.11.100/304 MAC/IP
252 * 145.1.1.3 100 I
253 2:145.1.1.3:11::0::e4:23:3c:9d:73:41::100.11.11.110/304 MAC/IP
254 * 145.1.1.3 100 I
255 2:145.1.1.6:11::0::00:00:00:01:01:01::100.11.11.1/304 MAC/IP
256 * 145.1.1.6 100 I
257 2:145.1.1.6:11::0::68:22:8e:1d:ab:b6::100.11.11.3/304 MAC/IP
258 * 145.1.1.6 100 I
259 2:145.1.1.6:11::0::88:30:37:6e:a1:c3::100.11.11.10/304 MAC/IP
260

```

```

261 * 145.1.1.6 100 I
262 2:145.1.1.6:11::0::88:30:37:6e:a1:db::100.11.11.20/304 MAC/IP
263 * 145.1.1.6 100 I
264 3:145.1.1.3:11::0::145.1.1.3/248 IM
265 * 145.1.1.3 100 I
266 3:145.1.1.6:11::0::145.1.1.6/248 IM
267 * 145.1.1.6 100 I

```

Check same outputs for ACX-PE2. In this case Host-1 and Host-2 MAC addresses are learned through the CE-Ring directly (“classic” data-plane l2 mac learning). Host-3 MAC is only communicated through remote ACX-PE3, via evpn bgp mac/mac-ip learning.

On ACX7100 PE2:

```

1 [edit]
2 aris@acx7100-210# show interfaces et-0/0/10 | display inheritance no-comments
3 flexible-vlan-tagging;
4 mtu 9000;
5 encapsulation flexible-ethernet-services;
6 unit 10 {
7     encapsulation vlan-bridge;
8     vlan-id 10;
9 }
10 unit 100 {
11     encapsulation vlan-bridge;
12     vlan-id 100;
13 }
14
15 ### PE2 ERPS State
16
17 [edit]
18 aris@acx7100-210# run show protection-group ethernet-ring node-state detail
19 Ethernet-Ring name : erp1
20 APS State : idle
21 Event : NR-RB
22 Ring Protection Link Owner : No
23 Wait to Restore Timer : disabled
24 Wait to Block Timer : disabled
25 Guard Timer : disabled
26 Operation state : operational
27
28 [edit]
29 aris@acx7100-210# run show protection-group ethernet-ring interface detail
30
31 Ethernet ring port parameters for protection group erp1
32
33 Interface name : et-0/0/10
34 Control channel name : et-0/0/10.10
35 Interface direction : east
36 Ring Protection Link End : No
37 Signal Failure : Clear
38 Forward State : forwarding
39 Interface Admin State : IFF ready
40
41 Interface name : Unused
42 Control channel name : Unused
43 Interface direction : west
44 Ring Protection Link End : NA
45 Signal Failure : NA
46 Forward State : NA
47 Interface Admin State : NA
48
49 ### PE1 EVPN State
50
51 [edit]
52 aris@acx7100-210# run show evpn instance extensive evpnerps
53 Instance: evpnerps
54 Route Distinguisher: 145.1.1.6:11
55 VLAN ID: 100
56 Per-instance MAC route label: 16
57 Control word enabled
58 Duplicate MAC detection threshold: 5
59 Duplicate MAC detection window: 180
60 MAC database status Local Remote
61 MAC advertisements: 3 4
62 MAC+IP advertisements: 4 4
63 Default gateway MAC advertisements: 2 1
64 Number of local interfaces: 2 (2 up)

```

```

65 Interface name ESI Mode Status AC-Role
66 .local..53 00:00:00:00:00:00:00:00:00 single-homed Up Root
67 et-0/0/10.100 00:00:00:00:00:00:00:00 single-homed Up Root
68 Number of IRB interfaces: 1 (1 up)
69 Interface name VLAN VNI Status L3 context
70 irb.100 100 Up master
71 Number of protect interfaces: 0
72 Number of bridge domains: 1
73 VLAN Domain-ID Intfs/up IRB-intf Mode MAC-sync v4-SG-sync v6-SG-sync
74 100 1 1 irb.100 Extended Enabled Disabled Disabled
75 Number of neighbors: 2
76 Address MAC MAC+IP AD IM ES Leaf-label DCI-Peer Flow-label DT2U-SID DT2M-SID
77 145.1.1.3 2 2 0 1 0 NO
78 145.1.1.5 2 2 1 1 0 NO
79 Number of ethernet segments: 1
80 ESI: 05:00:00:fe:4c:00:00:00:64:00
81 State-Bitfield: 0x43
82 ESI Route Label: 18
83 ESI Refcount: 1
84 ESI Num Macs: 1, ESI Num SGDBs: 0
85 Token-Route NH: 0
86 Number of Local interfaces: 1
87 Local interface: irb.100, Status: Up/Forwarding
88 Number of remote PEs connected: 1
89 Remote-PE MAC-label Aliasing-label Mode
90 145.1.1.5 17 0 all-active
91 DF Election Algorithm: MOD based
92 Designated forwarder: DF not elected yet
93 Last designated forwarder update: Never
94 Advertised split horizon label: 202
95 SMET Forwarding: Disabled
96 RIB Table-ID: 184549386, Kernel Table-ID: 53, Kernel Table-Generation: 0
97 EVPN instance flags: 0x801810000
98 RTT Update Timestamp: Jan 17 12:36:18.882 2025
99 L2ALD state change Timestamp: Jan 17 12:36:19.626 2025
100 Core-Isolation change TS: Apr 10 14:19:18.845 2025, Core-Isolated: N
101 Last Core-Isolation Change Reason: evpn-peer-transition
102
103 [edit]
104 aris@acx7100-210#
105
106 ### Local Host MAC addressses are learned through L2 direct MAC learning, since CE3 direct ring
107 uplink is non-blocked in non-failure condition.
108
109 [edit]
110 aris@acx7100-210# run show mac-vrf forwarding mac-table instance evpnerps
111
112 MAC flags (S - static MAC, D - dynamic MAC, L - locally learned, P - Persistent static, C -
113 Control MAC
114 SE - statistics enabled, NM - non configured MAC, R - remote PE MAC, O - ovsdb MAC
115 GBP - group based policy, B - Blocked MAC)
116
117 Ethernet switching table : 6 entries, 6 learned
118 Routing instance : evpnerps
119 Vlan MAC MAC Age GBP Logical NH MAC RTR
120 name address flags Tag interface Index property ID
121 vl100 00:00:00:01:01:01 DCP - 8818 8818
122 vl100 68:22:8e:1d:9b:a6 DCP - 8819 8819
123 vl100 88:30:37:6e:a1:c3 D - et-0/0/10.100 0 0 ### Direct dynamic mac learning of host-1 mac
124 vl100 88:30:37:6e:a1:db D - et-0/0/10.100 0 0 ### Direct dynamic mac learning of host-2 mac
125 vl100 ac:78:d1:22:3b:82 DC - 8821 8821 ### Control-plane dynamic mac learning of host-3 mac
126 vl100 e4:23:3c:9d:73:41 DC - 8821 8821
127
128 [edit]
129
130 ### MAC-IP routes
131
132 aris@acx7100-210# run show mac-vrf forwarding mac-ip-table instance evpnerps
133
134 MAC IP flags (S - Static, D - Dynamic, L - Local , R - Remote, Lp - Local Proxy,
135 Rp - Remote Proxy, K - Kernel, RT - Dest Route, (N)AD - (Not) Advt to remote,
136 RE - Re-ARP/ND, RO - Router, OV - Override, Ur - Unresolved, B - Blocked,
137 RTS - Dest Route Skipped, RGW - Remote Gateway, GBP - Group Based Policy,
138 RTF - Dest Route Forced, SC - Static Config, P - Probe, NLC - No Local Config,
139 LD - Local Down)
140 Routing instance : evpnerps
141 Bridging domain : vl100
142 IP MAC Flags GBP Logical Active
143 address address Tag Interface source
144 100.11.11.1 00:00:00:01:01:01 S,K irb.100 05:00:00:fe:4c:00:00:00:64:00
145 100.11.11.2 68:22:8e:1d:9b:a6 SR,K,RT 145.1.1.5

```

```

144 100.11.11.3 68:22:8e:1d:ab:b6 S,K irb.100
145 100.11.11.10 88:30:37:6e:a1:c3 DL,K,RT,AD et-0/0/10.100
146 100.11.11.20 88:30:37:6e:a1:db DL,K,RT,AD et-0/0/10.100
147 100.11.11.100 ac:78:d1:22:3b:82 SR,K,RT,RGw 145.1.1.3
148 100.11.11.110 e4:23:3c:9d:73:41 DR,K,RT 145.1.1.3
149
150 aris@acx7100-210# run show route forwarding-table table evpnerps | find 88:30:37:6e:a1:c3 ###
    LOCAL HOST-1
151 88:30:37:6e:a1:c3/48 user 0 ucst 10016 1 et-0/0/10.100
152 88:30:37:6e:a1:db/48 user 0 ucst 10016 1 et-0/0/10.100
153 <output ommited...>
154
155 ### Similar outputs for LOCAL HOST2 MAC: 88:30:37:6e:a1:db (omitted for brevity)####
156
157 [edit]
158 aris@acx7100-210# run show route forwarding-table table evpnerps | find e4:23:3c:9d:73:41 ###
    REMOTE HOST-3
159 e4:23:3c:9d:73:41/48 user 0 indr 8821 1
160         ulst 8523 1
161         sftw Push 17, Push 100003(top) 8510 1 et-0/0/0.0
162         10.100.0.45 ucst 1642 1 et-0/0/0.0
163         sftw Push 17, Push 100003(top) 8511 1 et-0/0/7.0
164         10.100.0.39 ucst 1001 1 et-0/0/7.0
165 <output ommited...>
166
167 [edit]
168 aris@acx7100-210# run show route table mpls.0 label 100003 ### explicit-null is configured for
    SR-MPLS
169
170 mpls.0: 48 destinations, 49 routes (48 active, 0 holddown, 0 hidden)
171 + = Active Route, - = Last Active, * = Both
172
173 100003 *[L-OSPF/6/5] 01:44:10, metric 15
174     > to 10.100.0.45 via et-0/0/0.0, Swap 100003
175     to 10.100.0.39 via et-0/0/7.0, Swap 100003
176
177 [edit]
178 aris@acx7100-210# run show route table mpls.0 label 17
179
180 mpls.0: 48 destinations, 49 routes (48 active, 0 holddown, 0 hidden)
181 + = Active Route, - = Last Active, * = Both
182
183 17 *[EVPN/7] 37w2d 07:35:42, routing-instance evpnerps, route-type Ingress-MAC, vlan-id 100
184     to table evpnerps.evpn-mac.0
185     [EVPN/7] 37w2d 07:35:42, routing-instance evpnerps, route-type Ingress-MAC, vlan-
        id 100
        to table evpnerps.evpn-mac.0
186
187 [edit]
188 aris@acx7100-210# run show route advertising-protocol bgp 145.1.1.1 table evpnerps.evpn.0
189
190 evpnerps.evpn.0: 20 destinations, 31 routes (20 active, 0 holddown, 0 hidden)
191 Prefix Nexthop MED Lclpref AS path
192 2:145.1.1.6:11::0::00:00:00:01:01:01/304 MAC/IP
193 * Self 100 I
194 2:145.1.1.6:11::0::68:22:8e:1d:ab:b6/304 MAC/IP
195 * Self 100 I
196 2:145.1.1.6:11::0::88:30:37:6e:a1:c3/304 MAC/IP
197 * Self 100 I
198 2:145.1.1.6:11::0::88:30:37:6e:a1:db/304 MAC/IP
199 * Self 100 I
200 2:145.1.1.6:11::0::00:00:00:01:01:01::100.11.11.1/304 MAC/IP
201 * Self 100 I ### VGA-MAC
202 2:145.1.1.6:11::0::68:22:8e:1d:ab:b6::100.11.11.3/304 MAC/IP
203 * Self 100 I ### IRB ACX-PE2 System MAC
204 2:145.1.1.6:11::0::88:30:37:6e:a1:c3::100.11.11.10/304 MAC/IP
205 * Self 100 I
206 2:145.1.1.6:11::0::88:30:37:6e:a1:db::100.11.11.20/304 MAC/IP
207 * Self 100 I
208 3:145.1.1.6:11::0::145.1.1.6/248 IM
209 * Self 100 I
210
211 [edit]
212 aris@acx7100-210# run show route receive-protocol bgp 145.1.1.1 table evpnerps.evpn.0
213
214 evpnerps.evpn.0: 20 destinations, 31 routes (20 active, 0 holddown, 0 hidden)
215 Prefix Nexthop MED Lclpref AS path
216 1:145.1.1.5:0::050000fe4c0000006400::FFFF:FFFF/192 AD/ESI
217 * 145.1.1.5 100 I
218 2:145.1.1.3:11::0::ac:78:d1:22:3b:82/304 MAC/IP
219 * 145.1.1.3 100 I
220

```

```

221 2:145.1.1.3:11::0::e4:23:3c:9d:73:41/304 MAC/IP
222 * 145.1.1.3 100 I
223 2:145.1.1.5:11::0::00:00:00:01:01:01/304 MAC/IP
224 * 145.1.1.5 100 I
225 2:145.1.1.5:11::0::68:22:8e:1d:9b:a6/304 MAC/IP
226 * 145.1.1.5 100 I
227 2:145.1.1.3:11::0::ac:78:d1:22:3b:82::100.11.11.100/304 MAC/IP
228 * 145.1.1.3 100 I
229 2:145.1.1.3:11::0::e4:23:3c:9d:73:41::100.11.11.110/304 MAC/IP
230 * 145.1.1.3 100 I
231 2:145.1.1.5:11::0::00:00:00:01:01:01::100.11.11.1/304 MAC/IP
232 * 145.1.1.5 100 I
233 2:145.1.1.5:11::0::68:22:8e:1d:9b:a6::100.11.11.2/304 MAC/IP
234 * 145.1.1.5 100 I
235 3:145.1.1.3:11::0::145.1.1.3/248 IM
236 * 145.1.1.3 100 I
237 3:145.1.1.5:11::0::145.1.1.5/248 IM
238 * 145.1.1.5 100 I

```

Let us also check the EVPN status of the tail-end ACX7100-PE3. Only Host-3 MAC is directly learned and all others through evpn bgp control-plane learning, as expected. It worth noting also that in normal operation MAC-IP routes are following the forwarding path to ACX-PE2, with CE link towards the ERPS ring in forwarding-state! No blackholing indeed!

On Remote ACX7100 PE3:

```

1 aris@acx7100-211# run show evpn instance extensive evpnerps
2 Instance: evpnerps
3 Route Distinguisher: 145.1.1.3:11
4 VLAN ID: 100
5 Per-instance MAC route label: 16
6 Control word enabled
7 Duplicate MAC detection threshold: 5
8 Duplicate MAC detection window: 180
9 MAC database status Local Remote
10 MAC advertisements: 2 6
11 MAC+IP advertisements: 2 5
12 Default gateway MAC advertisements: 1 0
13 Number of local interfaces: 2 (2 up)
14 Interface name ESI Mode Status AC-Role
15 .local..58 00:00:00:00:00:00:00:00:00 single-homed Up Root
16 et-0/0/12.100 00:00:00:00:00:00:00:00:00 single-homed Up Root
17 Number of IRB interfaces: 1 (1 up)
18 Interface name VLAN VNI Status L3 context
19 irb.100 100 Up master
20 Number of protect interfaces: 0
21 Number of bridge domains: 1
22 VLAN Domain-ID Intfs/up IRB-intf Mode MAC-sync v4-SG-sync v6-SG-sync
23 100 1 1 irb.100 Extended Enabled Disabled Disabled
24 Number of neighbors: 2
25 Address MAC MAC+IP AD IM ES Leaf-label Remote-DCI-Peer Flow-label
26 145.1.1.5 2 2 1 1 0 NO
27 145.1.1.6 4 4 1 1 0 NO
28 Number of ethernet segments: 1
29 ESI: 05:00:00:fe:4c:00:00:00:64:00
30 Local interface: irb.100, Status: Up/Forwarding
31 Number of remote PEs connected: 2
32 Remote-PE MAC-label Aliasing-label Mode
33 145.1.1.5 17 0 all-active
34 145.1.1.6 17 0 all-active
35 SMET Forwarding: Disabled
36
37 [edit]
38 aris@acx7100-211# run show mac-vrf forwarding mac-table instance evpnerps
39
40 MAC flags (S - static MAC, D - dynamic MAC, L - locally learned, P - Persistent static, C -
   Control MAC
41 SE - statistics enabled, NM - non configured MAC, R - remote PE MAC, O - ovsdb MAC
42 GBP - group based policy)
43
44 Ethernet switching table : 6 entries, 6 learned
45 Routing instance : evpnerps
46 Vlan MAC MAC Age GBP Logical NH RTR
47 name address flags Tag interface Index ID
48 vl100 00:00:00:01:01:01 DC - 51967 51967
49 vl100 68:22:8e:1d:9b:a6 DCP - 51729 51729
50 vl100 68:22:8e:1d:ab:b6 DCP - 51943 51943

```

```

51 vl100 88:30:37:6e:a1:c3 DC - 51943 51943
52 vl100 88:30:37:6e:a1:db DC - 51943 51943
53 vl100 e4:23:3c:9d:73:41 D - et-0/0/12.100 0 0
54
55 ### MAC-IP routes for Host-1,2 pointing towards ACX7K-PE2 (145.1.1.6 source)
56
57 [edit]
58 aris@acx7100-211# run show mac-vrf forwarding mac-ip-table instance evpnerps
59
60 MAC IP flags (S - Static, D - Dynamic, L - Local , R - Remote, Lp - Local Proxy,
61                Rp - Remote Proxy, K - Kernel, RT - Dest Route, (N)AD - (Not) Advt to remote,
62                RE - Re-ARP/ND, RO - Router, OV - Override, Ur - Unresolved,
63                RTS - Dest Route Skipped, RGW - Remote Gateway, GBP - Group Based Policy,
64                RTF - Dest Route Forced)
65 Routing instance : evpnerps
66 Bridging domain : vl100
67 IP MAC Flags GBP Logical Active
68 address address Tag Interface source
69 100.11.11.1 00:00:00:01:01:01 SR,K 05:00:00:fe:4c:00:00:00:64:00
70 100.11.11.2 68:22:8e:1d:9b:a6 SR,K,RT 145.1.1.5
71 100.11.11.3 68:22:8e:1d:ab:b6 SR,K,RT 145.1.1.6
72 100.11.11.10 88:30:37:6e:a1:c3 DR,K,RT 145.1.1.6
73 100.11.11.20 88:30:37:6e:a1:db DR,K,RT 145.1.1.6
74 100.11.11.100 ac:78:d1:22:3b:82 S,K irb.100
75 100.11.11.110 e4:23:3c:9d:73:41 DL,K,RT,AD et-0/0/12.100
76
77 [edit]
78 aris@acx7100-211# run show route forwarding-table table evpnerps | find 88:30:37:6e:a1:c3
79 88:30:37:6e:a1:c3/48 user 0 indr 51943 1
80                ulst 51333 1
81                sftw Push 17, Push 100006(top) 51306 1 ae0.0
82                10.100.0.11 ucst 1000 1 ae0.0
83                sftw Push 17, Push 100006(top) 51307 1 et-0/0/0.0
84                10.100.0.61 ucst 1002 1 et-0/0/0.0
85 <output omitted for brevity>
86
87 [edit]
88 aris@acx7100-211# run show route table mpls.0 label 100006 ### explicit-null is configured for
89 SR-MPLS
90 mpls.0: 54 destinations, 54 routes (54 active, 0 holddown, 0 hidden)
91 + = Active Route, - = Last Active, * = Both
92
93 100006 *[L-OSPF/6/5] 03:29:45, metric 15
94         > to 10.100.0.11 via ae0.0, Swap 100006
95         to 10.100.0.61 via et-0/0/0.0, Swap 100006
96
97 [edit]
98 aris@acx7100-211# run show route table mpls.0 label 17
99
100 mpls.0: 54 destinations, 54 routes (54 active, 0 holddown, 0 hidden)
101 + = Active Route, - = Last Active, * = Both
102
103 17 *[EVPN/7] 37w2d 09:21:21, routing-instance evpnerps, route-type Ingress-MAC, vlan-id 100
104     to table evpnerps.evpn-mac.0
105
106 [edit]
107 aris@acx7100-211# run show route advertising-protocol bgp 145.1.1.1 table evpnerps.evpn.0
108
109 evpnerps.evpn.0: 21 destinations, 37 routes (21 active, 0 holddown, 0 hidden)
110 Prefix Nexthop MED Lclpref AS path
111 2:145.1.1.3:11::0::ac:78:d1:22:3b:82/304 MAC/IP
112 * Self 100 I
113 2:145.1.1.3:11::0::e4:23:3c:9d:73:41/304 MAC/IP
114 * Self 100 I ### sourcing Host-3 MAC towards
115 2:145.1.1.3:11::0::ac:78:d1:22:3b:82::100.11.11.100/304 MAC/IP
116 * Self 100 I
117 2:145.1.1.3:11::0::e4:23:3c:9d:73:41::100.11.11.110/304 MAC/IP
118 * Self 100 I
119 3:145.1.1.3:11::0::145.1.1.3/248 IM
120 * Self 100 I
121
122 ### Receiving via evpn bgp control plane learning Host1,2 mac/ip routes
123
124 [edit]
125 aris@acx7100-211# run show route receive-protocol bgp 145.1.1.1 table evpnerps.evpn.0
126
127 evpnerps.evpn.0: 21 destinations, 37 routes (21 active, 0 holddown, 0 hidden)
128 Prefix Nexthop MED Lclpref AS path
129 1:145.1.1.5:0::050000fe4c0000006400::FFFF:FFFF/192 AD/ESI
130 * 145.1.1.5 100 I

```

```

131 1:145.1.1.6:0::050000fe4c0000006400::FFFF:FFFF/192 AD/ESI
132 * 145.1.1.6 100 I
133 2:145.1.1.5:11::00:00:00:01:01:01/304 MAC/IP
134 * 145.1.1.5 100 I
135 2:145.1.1.5:11::00:68:22:8e:1d:9b:a6/304 MAC/IP
136 * 145.1.1.5 100 I
137 2:145.1.1.6:11::00:00:00:01:01:01/304 MAC/IP
138 * 145.1.1.6 100 I
139 2:145.1.1.6:11::00:68:22:8e:1d:ab:b6/304 MAC/IP
140 * 145.1.1.6 100 I
141 2:145.1.1.6:11::00:88:30:37:6e:a1:c3/304 MAC/IP
142 * 145.1.1.6 100 I
143 2:145.1.1.6:11::00:88:30:37:6e:a1:db/304 MAC/IP
144 * 145.1.1.6 100 I
145 2:145.1.1.5:11::00:00:00:01:01:01::100.11.11.1/304 MAC/IP
146 * 145.1.1.5 100 I
147 2:145.1.1.5:11::00:68:22:8e:1d:9b:a6::100.11.11.2/304 MAC/IP
148 * 145.1.1.5 100 I
149 2:145.1.1.6:11::00:00:00:01:01:01::100.11.11.1/304 MAC/IP
150 * 145.1.1.6 100 I
151 2:145.1.1.6:11::00:68:22:8e:1d:ab:b6::100.11.11.3/304 MAC/IP
152 * 145.1.1.6 100 I
153 2:145.1.1.6:11::00:88:30:37:6e:a1:c3::100.11.11.10/304 MAC/IP
154 * 145.1.1.6 100 I
155 2:145.1.1.6:11::00:88:30:37:6e:a1:db::100.11.11.20/304 MAC/IP
156 * 145.1.1.6 100 I
157 3:145.1.1.5:11::00::145.1.1.5/248 IM
158 * 145.1.1.5 100 I
159 3:145.1.1.6:11::00::145.1.1.6/248 IM
160 * 145.1.1.6 100 I

```

Any-to-any forwarding connectivity is feasible in normal ERPS-to-EVPN operation. E.g. using rapid pings we can verify from Host-1 connectivity to local Host-2 and remote Host-3 respectively. Traffic is forwarded towards ACX7100-PE2 and then through EVPN-MPLS overlay transport to ACX7100-PE3 tail-end router, to L2 hand-off towards remote Host-3 (100.11.11.110).

Hence performing rapid ping in upstream direction, from Host-1:

```

1  ### LOCAL HOST-1 in L2 RING SIDE #####
2
3  {master:0}[edit]
4  aris@qfx5120-269# show routing-instances
5  VR1 {
6      instance-type virtual-router;
7      routing-options {
8          static {
9              route 0.0.0.0/0 next-hop 100.11.11.1;
10         }
11     }
12     interface et-0/0/4.1;
13 }
14
15 {master:0}[edit]
16 aris@qfx5120-269# show interfaces et-0/0/4
17 flexible-vlan-tagging;
18 mtu 9000;
19 encapsulation flexible-ethernet-services;
20 unit 1 {
21     vlan-id 100;
22     family inet {
23         address 100.11.11.10/24;
24     }
25 }
26
27 {master:0}[edit]
28 aris@qfx5120-269# run show arp vpn VR1
29 MAC Address Address Name Interface Flags
30 00:00:00:01:01:01 100.11.11.1 100.11.11.1 et-0/0/4.1 none
31 68:22:8e:1d:9b:a6 100.11.11.2 100.11.11.2 et-0/0/4.1 none
32 68:22:8e:1d:ab:b6 100.11.11.3 100.11.11.3 et-0/0/4.1 none
33 88:30:37:6e:a1:db 100.11.11.20 100.11.11.20 et-0/0/4.1 none
34 ac:78:d1:22:3b:82 100.11.11.100 100.11.11.100 et-0/0/4.1 none
35 e4:23:3c:9d:73:41 100.11.11.110 100.11.11.110 et-0/0/4.1 none
36 Total entries: 6
37
38 {master:0}[edit]
39 aris@qfx5120-269# run ping rapid count 5 routing-instance VR1 100.11.11.20

```


[illegible]

Rapid Ping traffic seems to ingress/egress from ACX-PE2 (aka acx7100-210), as expected

```

1  Interface Link Input bytes (bps) Output bytes (bps) Description
2  ...
3  et-0/0/0 Up 61630741209 (6017992) 370084082890 (6077480) to-acx7100-212
4  ...
5  et-0/0/7 Up 15540974225 (4152) 15455894758 (4872) to-acx7100-209
6  ...
7  et-0/0/10 Up 1500498543 (6028472) 1177382146 (6028472) to-qfx5120-75

```

And acx7100-209:

```

1      Interface Link Input bytes (bps) Output bytes (bps) Description
2      ...
3      et-0/0/0 Up 60306252528 (10168) 358647747614 (11336) to-acx7100-5
4      ...
5      et-0/0/7 Up 15456011774 (4664) 15541094324 (7272) to-acx7100-10
6      ...
7      et-0/0/10 Up 637614175 (0) 314929576 (0)

```

Performing similar rapid pings in downstream direction, from Host-3; REMOTE HOST-3 behind EVPN network side (aka qfx5120-272 VR1 RI in L2 RING SIDE):

```

1 {master:0}[edit]
2 aris@qfx5120-272# show routing-instances
3 VR1 {
4     instance-type virtual-router;
5     interface et-0/0/51.100;
6 }
7
8 {master:0}[edit]
9 aris@qfx5120-272# show interfaces et-0/0/51
10 description acx7100-211;
11 flexible-vlan-tagging;
12 mtu 9000;
13 encapsulation flexible-ethernet-services;
14 unit 100 {
15     vlan-id 100;
16     family inet {
17         address 100.11.11.110/24;
18     }
19 }
20
21 {master:0}[edit]
22 aris@qfx5120-272# run show arp | match 100.11.11
23 88:30:37:6e:a1:c3 100.11.11.10 100.11.11.10 et-0/0/51.100 none
24 88:30:37:6e:a1:db 100.11.11.20 100.11.11.20 et-0/0/51.100 none
25 ac:78:d1:22:3b:82 100.11.11.100 100.11.11.100 et-0/0/51.100 none
26
27 {master:0}[edit]
28 aris@qfx5120-272# run ping rapid count 5 routing-instance VR1 100.11.11.10
29 PING 100.11.11.10 (100.11.11.10): 56 data bytes
30 !!!!!

```


[illegible]

Rapid Ping traffic seems to ingress/egress from ACX-PE3 (aka acx7100-211), as expected.

acx7100-211 (remote PE-3):

```

1 Interface Link Input bytes (bps) Output bytes (bps) Description
2
3 ae0 Up 149489244848 (5521200) 240721267579709 (111246144)
4 ...
5 et-0/0/0 Up 358651943115 (10144) 60307002259 (10000) to-acx7100-209
6 et-0/0/6 Up 67047167073 (5518240) 127839255912257 (4987136) to-acx7100-212
7 et-0/0/7 Up 82442077775 (2960) 112882011667452 (106259008) to-acx7100-212
8 ...
9 et-0/0/12 Up 294823654849603 (111184184) 158578310238037 (62808168) qfx5120-272

```

1.6 ERPS-to-EVPN In Action @Failure Condition, Fast Convergence

Let us now create some indicative failure conditions and check the effect on our rapid ping bidirectional traffic.

Failure Condition-1: Disable RPL owner link (CE2-PE1), or a non-RPL link (e.g. CE3-PE2) and check traffic forwarding fast convergence.

Then enable failed links and check whether the traffic converges back again to normal operation path (after WTR period elapses), with minimal packet loss. Check the new MAC learning stanza on ERPS L2 ring and EVPN PEs.

Setting laser-off on ACX7100-209 to simulate CE2-PE1 RPL owner link failure:

```

1 aris@acx7100-209:pfe> test picd optics fpc_slot 0 pic_slot 0 port 10 cmd laser-off
2
3 PICD optics laser-off
4 xcvr-0/0/10: Switched OFF laser on all channels
5
6 aris@acx7100-209:pfe> exit
7
8 [vrf:none] aris@acx7100-209:~$ exit
9 exit
10
11 aris@acx7100-209> show interfaces et-0/0/10 terse
12 Interface Admin Link Proto Local Remote
13 et-0/0/10 up down
14 et-0/0/10.10 up down ethernet-switching
15 et-0/0/10.100 up down ethernet-switching
16 et-0/0/10.32767 up down multiservice
17
18 ### ERPS State
19
20 aris@acx7100-209> show protection-group ethernet-ring node-state detail
21 Ethernet-Ring name : erp1

```

```

22 APS State : protected
23 Event : local SF ### SF Condition detected
24 Ring Protection Link Owner : No
25 Wait to Restore Timer : disabled
26 Wait to Block Timer : disabled
27 Guard Timer : disabled
28 Operation state : operational
29
30 aris@acx7100-209> show protection-group ethernet-ring interface detail
31
32 Ethernet ring port parameters for protection group erp1
33
34 Interface name : et-0/0/10
35 Control channel name : et-0/0/10.10
36 Interface direction : east
37 Ring Protection Link End : No
38 Signal Failure : Set ### Set SF condition
39 Forward State : discarding ### Discarding/BLK condition
40 Interface Admin State : IFF ready
41
42 Interface name : Unused
43 Control channel name : Unused
44 Interface direction : west
45 Ring Protection Link End : NA
46 Signal Failure : NA
47 Forward State : NA
48 Interface Admin State : NA

```

QFX-CE2 RPL Owner Status after failure condition:

```

1 aris@qfx5120-274# run show protection-group ethernet-ring node-state detail
2 Ethernet-Ring name : erp1
3 APS State : protected ### protected state under failure condition
4 Event : local SF ###
5 Ring Protection Link Owner : Yes
6 Wait to Restore Timer : disabled
7 Wait to Block Timer : disabled
8 Guard Timer : disabled
9 Operation state : operational
10
11 {master:0}[edit]
12 aris@qfx5120-274# run show protection-group ethernet-ring interface detail
13
14 Ethernet ring port parameters for protection group erp1
15
16 Interface name : et-0/0/49
17 Control channel name : et-0/0/49.10
18 Interface direction : east
19 Ring Protection Link End : No
20 Signal Failure : Clear
21 Forward State : forwarding
22 Interface Admin State : IFF ready
23
24 Interface name : et-0/0/48
25 Control channel name : et-0/0/48.10
26 Interface direction : west
27 Ring Protection Link End : Yes
28 Signal Failure : Set ### Set SF condition
29 Forward State : discarding ### Discarding/BLK condition
30 Interface Admin State : IFF ready

```

0% packet loss on rapid pings for both directions.

Recovering the RPL owner link failure, i.e. cmd laser-on

```

1 aris@acx7100-209> show protection-group ethernet-ring node-state detail
2 Ethernet-Ring name : erp1
3 APS State : pending
4 Event : NR
5 Ring Protection Link Owner : No
6 Wait to Restore Timer : disabled
7 Wait to Block Timer : disabled
8 Guard Timer : disabled
9 Operation state : operational
10
11 aris@acx7100-209> show protection-group ethernet-ring interface detail
12
13 Ethernet ring port parameters for protection group erp1

```

```

14
15 Interface name : et-0/0/10
16 Control channel name : et-0/0/10.10
17 Interface direction : east
18 Ring Protection Link End : No
19 Signal Failure : Clear
20 Forward State : forwarding
21 Interface Admin State : IFF ready
22
23 Interface name : Unused
24 Control channel name : Unused
25 Interface direction : west
26 Ring Protection Link End : NA
27 Signal Failure : NA
28 Forward State : NA
29 Interface Admin State : NA
30
31 {master:0}[edit]

```

On the QFX:

```

1 aris@qfx5120-274# run show protection-group ethernet-ring node-state detail
2 Ethernet-Ring name : erp1
3 APS State : pending
4 Event : WTR running
5 Ring Protection Link Owner : Yes
6 Wait to Restore Timer : running (time to expire: 186 sec) ### WTR timer running
7 Wait to Block Timer : disabled
8 Guard Timer : disabled
9 Operation state : operational
10
11 {master:0}[edit]
12 aris@qfx5120-274# run show protection-group ethernet-ring interface detail
13
14 Ethernet ring port parameters for protection group erp1
15
16 Interface name : et-0/0/49
17 Control channel name : et-0/0/49.10
18 Interface direction : east
19 Ring Protection Link End : No
20 Signal Failure : Clear
21 Forward State : forwarding
22 Interface Admin State : IFF ready
23
24 Interface name : et-0/0/48
25 Control channel name : et-0/0/48.10
26 Interface direction : west
27 Ring Protection Link End : Yes
28 Signal Failure : Clear ### Clear SF condition
29 Forward State : discarding ### Still Discarding/BLK
30 Interface Admin State : IFF ready

```

After WTR timer elapsed, everything back to normal

```

1 aris@qfx5120-274# run show protection-group ethernet-ring node-state detail
2 Ethernet-Ring name : erp1
3 APS State : idle
4 Event : NR-RB
5 Ring Protection Link Owner : Yes
6 Wait to Restore Timer : disabled
7 Wait to Block Timer : disabled
8 Guard Timer : disabled
9 Operation state : operational
10
11 {master:0}[edit]
12 aris@qfx5120-274# run show protection-group ethernet-ring interface detail
13
14 Ethernet ring port parameters for protection group erp1
15
16 Interface name : et-0/0/49
17 Control channel name : et-0/0/49.10
18 Interface direction : east
19 Ring Protection Link End : No
20 Signal Failure : Clear
21 Forward State : forwarding
22 Interface Admin State : IFF ready
23
24 Interface name : et-0/0/48
25 Control channel name : et-0/0/48.10

```

```

26 Interface direction : west
27 Ring Protection Link End : Yes
28 Signal Failure : Clear
29 Forward State : discarding ### RPL link Discarding/BLK
30 Interface Admin State : IFF ready

```

0% packet loss on rapid pings for both directions after RPL owner link failure restoration. Setting laser-off on ACX-PE2 (aka ACX7100-210) to simulate CE3-PE2 non-RPL link failure. Traffic flowing from ACX7K-PE2 uplink towards remote ACX7K-PE3 (normal operation)

On acx7100-210:

```

1 ...
2 et-0/0/0 Up 66972020926 (6195080) 376064410095 (6256312) to-acx7100-212
3 ...

```

On acx7100-209:

```

1 ...
2 et-0/0/0 Up 60461226443 (10064) 359419018724 (11336) to-acx7100-211
3 ...

```

```

1 aris@acx7100-210:pfe> test picd optics fpc_slot 0 pic_slot 0 port 10 cmd laser-off
2
3 PICD optics laser-off
4 xcvr-0/0/10: Switched OFF laser on all channels
5
6 aris@acx7100-210:pfe> exit
7
8 [vrf:none] aris@acx7100-210:~$ exit
9 Exit
10
11 aris@acx7100-210> show interfaces terse et-0/0/10
12 Interface Admin Link Proto Local Remote
13 et-0/0/10 up down
14 et-0/0/10.10 up down ethernet-switching
15 et-0/0/10.100 up down ethernet-switching
16 et-0/0/10.32767 up down multiservice
17
18 ### ERPS State
19
20 aris@acx7100-210> show protection-group ethernet-ring node-state detail
21 Ethernet-Ring name : erp1
22 APS State : protected
23 Event : local SF
24 Ring Protection Link Owner : No
25 Wait to Restore Timer : disabled
26 Wait to Block Timer : disabled
27 Guard Timer : disabled
28 Operation state : operational
29
30 aris@acx7100-210> show protection-group ethernet-ring interface detail
31
32 Ethernet ring port parameters for protection group erp1
33
34 Interface name : et-0/0/10
35 Control channel name : et-0/0/10.10
36 Interface direction : east
37 Ring Protection Link End : No
38 Signal Failure : Set ### Set SF condition
39 Forward State : discarding ### Transition from Forwarding to Discarding/BLK!
40 Interface Admin State : IFF ready
41
42 Interface name : Unused
43 Control channel name : Unused
44 Interface direction : west
45 Ring Protection Link End : NA
46 Signal Failure : NA
47 Forward State : NA
48 Interface Admin State : NA

```

QFX-CE2 (qfx5120-274) RPL Owner after failure condition

```

1 aris@qfx5120-274> show protection-group ethernet-ring node-state detail
2 Ethernet-Ring name : erp1

```

```

3 APS State : protected
4 Event : remote SF ### remote SF condition
5 Ring Protection Link Owner : Yes
6 Wait to Restore Timer : disabled
7 Wait to Block Timer : disabled
8 Guard Timer : disabled
9 Operation state : operational
10
11 {master:0}
12 aris@qfx5120-274> show protection-group ethernet-ring interface detail
13
14 Ethernet ring port parameters for protection group erp1
15
16 Interface name : et-0/0/49
17 Control channel name : et-0/0/49.10
18 Interface direction : east
19 Ring Protection Link End : No
20 Signal Failure : Clear
21 Forward State : forwarding
22 Interface Admin State : IFF ready
23
24 Interface name : et-0/0/48
25 Control channel name : et-0/0/48.10
26 Interface direction : west
27 Ring Protection Link End : Yes
28 Signal Failure : Clear
29 Forward State : forwarding ### Transition from Discarding/BLK to Forwarding!
30 Interface Admin State : IFF ready

```

QFX-CE3 (qfx5120-275) Non-RPL Owner after failure condition

```

1 aris@qfx5120-275> show protection-group ethernet-ring node-state detail
2 Ethernet-Ring name : erp1
3 APS State : protected
4 Event : local SF ### Local SF detected
5 Ring Protection Link Owner : No
6 Wait to Restore Timer : disabled
7 Wait to Block Timer : disabled
8 Guard Timer : disabled
9 Operation state : operational
10
11 {master:0}
12 aris@qfx5120-275> show protection-group ethernet-ring interface detail
13
14 Ethernet ring port parameters for protection group erp1
15
16 Interface name : et-0/0/48
17 Control channel name : et-0/0/48.10
18 Interface direction : east
19 Ring Protection Link End : No
20 Signal Failure : Set
21 Forward State : discarding ### Revert to Discarding/BLK!
22 Interface Admin State : IFF ready
23
24 Interface name : et-0/0/49
25 Control channel name : et-0/0/49.10
26 Interface direction : west
27 Ring Protection Link End : No
28 Signal Failure : Clear
29 Forward State : forwarding
30 Interface Admin State : IFF ready

```

Traffic now flowing “west” through ACX7K-PE1 unblink, under this failure condition

acx7100-209:

	Interface	Link	Input bytes (bps)	Output bytes (bps)	Description
1	et-0/0/0	Up	60598226640 (6488768)	359643088967 (6428608)	to-acx7100-5

Minimal packet loss on rapid pings for both directions (four rapids)

- CE1 → CE3 1562 packets transmitted, 1558 packets received, at the time of switchover
- CE3 → CE1 1562 packets transmitted, 1558 packets received, at the time of switchover

Recovering the CE3-PE2 link failure:

```

1  aris@acx7100-210:pfe> test picd optics fpc_slot 0 pic_slot 0 port 10 cmd laser-on
2
3  PICD optics laser-on
4  xcvr-0/0/10: Switched ON laser on all channels
5
6  aris@acx7100-210:pfe> exit
7
8  [vrf:none] aris@acx7100-210:~$ exit
9  exit
10
11 aris@acx7100-210> show interfaces terse et-0/0/10
12 Interface Admin Link Proto Local Remote
13 et-0/0/10 up up
14 et-0/0/10.10 up up ethernet-switching
15 et-0/0/10.100 up up ethernet-switching
16 et-0/0/10.32767 up up multiservice
17
18 aris@acx7100-210> show protection-group ethernet-ring node-state detail
19 Ethernet-Ring name : erp1
20 APS State : pending
21 Event : NR
22 Ring Protection Link Owner : No
23 Wait to Restore Timer : disabled
24 Wait to Block Timer : disabled
25 Guard Timer : disabled
26 Operation state : operational
27
28 aris@acx7100-210> show protection-group ethernet-ring interface detail
29
30 Ethernet ring port parameters for protection group erp1
31
32 Interface name : et-0/0/10
33 Control channel name : et-0/0/10.10
34 Interface direction : east
35 Ring Protection Link End : No
36 Signal Failure : Clear
37 Forward State : forwarding
38 Interface Admin State : IFF ready
39
40 Interface name : Unused
41 Control channel name : Unused
42 Interface direction : west
43 Ring Protection Link End : NA
44 Signal Failure : NA
45 Forward State : NA
46 Interface Admin State : NA

```

On QFX-CE2 (aka qfx5120-274) after failure recovery captures

```

1  aris@qfx5120-274> show protection-group ethernet-ring node detail
2  Ethernet-Ring name : erp1
3  APS State : pending
4  Event : WTR running
5  Ring Protection Link Owner : Yes
6  Wait to Restore Timer : running (time to expire: 173 sec) ### WTR timer running
7  Wait to Block Timer : disabled
8  Guard Timer : disabled
9  Operation state : operational
10
11 {master:0}
12 aris@qfx5120-274> show protection-group ethernet-ring interface detail
13
14 Ethernet ring port parameters for protection group erp1
15
16 Interface name : et-0/0/49
17 Control channel name : et-0/0/49.10
18 Interface direction : east
19 Ring Protection Link End : No
20 Signal Failure : Clear
21 Forward State : forwarding
22 Interface Admin State : IFF ready
23
24 Interface name : et-0/0/48
25 Control channel name : et-0/0/48.10
26 Interface direction : west
27 Ring Protection Link End : Yes
28 Signal Failure : Clear

```

```

29 Forward State : forwarding ### RPL link is still Forwarding, until WTR timer ellapses
30 Interface Admin State : IFF ready

```

On QFX-CE3 (aka qfx5120-275) after failure recovery captures

```

1  aris@qfx5120-275> show protection-group ethernet-ring node detail
2  Ethernet-Ring name : erp1
3  APS State : pending
4  Event : NR ### No Request condition
5  Ring Protection Link Owner : No
6  Wait to Restore Timer : disabled
7  Wait to Block Timer : disabled
8  Guard Timer : disabled
9  Operation state : operational
10
11 {master:0}
12 aris@qfx5120-275> show protection-group ethernet-ring interface detail
13
14 Ethernet ring port parameters for protection group erp1
15
16 Interface name : et-0/0/48
17 Control channel name : et-0/0/48.10
18 Interface direction : east
19 Ring Protection Link End : No
20 Signal Failure : Clear
21 Forward State : discarding ### Still Discarding/BLK, until WTR timer ellapses.
22 Interface Admin State : IFF ready
23
24 Interface name : et-0/0/49
25 Control channel name : et-0/0/49.10
26 Interface direction : west
27 Ring Protection Link End : No
28 Signal Failure : Clear
29 Forward State : forwarding
30 Interface Admin State : IFF ready

```

After WTR timer ellapsed, everything back to normal

QFX-CE2 (aka qfx5120-274) outputs:

```

1  aris@qfx5120-274> show protection-group ethernet-ring node detail
2  Ethernet-Ring name : erp1
3  APS State : idle
4  Event : NR-RB
5  Ring Protection Link Owner : Yes
6  Wait to Restore Timer : disabled
7  Wait to Block Timer : disabled
8  Guard Timer : disabled
9  Operation state : operational
10
11 {master:0}
12 aris@qfx5120-274> show protection-group ethernet-ring interface detail
13
14 Ethernet ring port parameters for protection group erp1
15
16 Interface name : et-0/0/49
17 Control channel name : et-0/0/49.10
18 Interface direction : east
19 Ring Protection Link End : No
20 Signal Failure : Clear
21 Forward State : forwarding
22 Interface Admin State : IFF ready
23
24 Interface name : et-0/0/48
25 Control channel name : et-0/0/48.10
26 Interface direction : west
27 Ring Protection Link End : Yes
28 Signal Failure : Clear
29 Forward State : discarding ### RPL link Discarding/Blocked again
30 Interface Admin State : IFF ready

```

Minimal rapid ping packet-loss upon restoration and traffic now forwarded from ACX7100-PE1 uplink again

- HOST-1 -> HOST-3 1845 packets transmitted, 1844 packets received.
- HOST-3 -> HOST-1 1845 packets transmitted, 1844 packets received.

acx7100-210:

```
1 Interface Link Input bytes (bps) Output bytes (bps) Description
2 ...
3 et-0/0/0 Up 67247431187 (6791032) 376463557637 (6733520) to-acx7100-212
```

Failure Condition-2: Fail (reboot) RPL node (CE2), or a non-RPL node (e.g. CE1) and check traffic forwarding fast convergence. Then after failed node comes online, check whether the traffic converges back again to normal operation path (after WTR period elapses), with minimal packet loss. Check the new MAC learning stanza on ERPS L2 ring and EVPN PEs.

```
1 ##Rebooting CE2 RPL node. Since Local HOST-1 (.10) is only hanging behind CE2 the rapid ping
  bidi test would be to/from HOST-2 (.20). Minimal rapid ping packet-loss on both node failure
  and online.
2 HOST-2 (.20)- CE1 --> HOST-3 (.110) 2668 packets transmitted, 2667 packets received
3 HOST-3 (.110) --> CE1-HOST-2 (.20) 2668 packets transmitted, 2667 packets received
4
5 ### Minimal rapid ping packet-loss upon restoration amd traffic now forwarded from ACX7100-PE1
  uplink again
6 HOST-1 (.10) - CE2 --> HOST-3 (.110) 3029 packets transmitted, 3025 packets received
7 HOST-3 (.110) --> HOST-1 (.10) - CE2 3029 packets transmitted, 3025 packets received
```

Note: You may also issue force-switch or manual-switch group commands, to toggle the state of an erps interface from forwarding to discarding. That results in traffic direction reversion, towards ACX-PE2. E.g. on QFX-CE2 (aka qfx5120-274):

```
1 aris@qfx5120-274> request protection-group ethernet-ring force-switch interface-name et-0/0/49
  group-name erp1
2 Local FS command is completed for pg (erp1) intf (et-0/0/49.10)
3 {master:0}
4 aris@qfx5120-274> show protection-group ethernet-ring interface
5
6 Ethernet ring port parameters for protection group erp1
7
8 Interface Control Channel Direction Forward State RPL End SF Admin State
9 et-0/0/49 et-0/0/49.10 east discarding No Clear IFF ready ### forced-switched
10 et-0/0/48 et-0/0/48.10 west forwarding Yes Clear IFF ready
11 {master:0}
12 aris@qfx5120-274> show protection-group ethernet-ring node-state
13 Ethernet ring APS State Event RPL Owner WTR Timer WTB Timer Guard Timer Operation state
14 erp1 FS local FS Yes disabled disabled disabled operational ### FS node state
```

If we check also on ACX7K-PE1 side

```
1 aris@acx7100-209> show protection-group ethernet-ring node-state
2 Ethernet ring APS State Event RPL Owner WTR Timer WTB Timer Guard Timer Operation state
3 erp1 FS remote FS No disabled disabled disabled operational ### Remote FS
4
5 aris@acx7100-209>
6
7 aris@acx7100-209> show protection-group ethernet-ring interface
8
9 Ethernet ring port parameters for protection group erp1
10
11 Interface Control Channel Direction Forward State RPL End SF Admin State
12 et-0/0/10 et-0/0/10.10 east forwarding No Clear IFF ready ### Reverted to Forwarding
13 Unused Unused west NA NA NA NA
```

Using the clear force-switch or manual-switch command, your restore back the normal operation mode. E.g. on QFX-CE2 (aka qfx5120-274):

```
1 aris@qfx5120-274> request protection-group ethernet-ring clear group-name erp1
2 CLEAR request is completed for pg (erp1)
3
4 ### Everything back to normal operation state
5
6 {master:0}
7 aris@qfx5120-274> show protection-group ethernet-ring interface
8
9 Ethernet ring port parameters for protection group erp1
10
```



```

11 Interface Control Channel Direction Forward State RPL End SF Admin State
12 et-0/0/49 et-0/0/49.10 east forwarding No Clear IFF ready
13 et-0/0/48 et-0/0/48.10 west discarding Yes Clear IFF ready
14
15 {master:0}
16 aris@qfx5120-274> show protection-group ethernet-ring node-state
17 Ethernet ring APS State Event RPL Owner WTR Timer WTB Timer Guard Timer Operation state
18 erp1 idle NR-RB Yes disabled disabled disabled operational

```

Same for ACX7K-PE1 side- everything back to normal:

```

1 aris@acx7100-209> show protection-group ethernet-ring node-state
2 Ethernet ring APS State Event RPL Owner WTR Timer WTB Timer Guard Timer Operation state
3 erp1 idle NR-RB No disabled disabled disabled operational
4
5 aris@acx7100-209> show protection-group ethernet-ring interface
6
7 Ethernet ring port parameters for protection group erp1
8
9 Interface Control Channel Direction Forward State RPL End SF Admin State
10 et-0/0/10 et-0/0/10.10 east forwarding No Clear IFF ready
11 Unused Unused west NA NA NA NA

```

1.7 Acknowledgements

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1.8 Useful links

- EVPN MAC-VRF Validation on ACX7000: <https://juniper.github.io/techposts/evpn-mac-vrf-validation-on-acx7000/article>
- EVPN E-LINE/E-LAN Service on ACX7000 Series: <https://www.juniper.net/documentation/us/en/software/nce/nce-501-evpn-e-line-e-lan-acx/nce-501-evpn-e-line-e-lan-acx.pdf>
- Understanding Ethernet Ring Protection Switching: <https://www.juniper.net/documentation/us/en/software/junos/high-availability/topics/topic-map/ethernet-ring-protection-switching-understanding.html>
- Configuring Ethernet Ring Protection Switching on Switches (CLI Procedure): https://www.juniper.net/documentation/en_US/junos/topics/task/configuration/ethernet-ring-protection-cli.html
- MPLS Applications User Guide: <https://www.juniper.net/documentation/us/en/software/junos/mpls/mpls.pdf>
- Day One: Configuring Segment Routing with JUNOS: https://www.juniper.net/documentation/en_US/day-one-books/DO_SegmentRouting.pdf

1.9 Glossary

- APS: Automatic Protection Switching
- BGP: Border Gateway Protocol
- BFD: Bidirectional Forwarding Detection
- BUM: Broadcast, Unko
- BUM: Broadcast, Unknown-unicast and Multicast (layer-2) Traffic
- CE: Customer Edge

- ELAN: Ethernet Local Area Network
- EPL: Ethernet Private Line
- ERPS: Ethernet Ring Protection Switching
- EVI: EVPN (MAC-VRF) Instance
- EVPL: Ethernet Virtual Private Line
- EVPN: Ethernet Virtual Private Network
- GRT: Global Routing Table
- IGP: Internal Gateway Protocol
- IPv4: Internet Protocol version 4
- IPv6: Internet Protocol version 6
- JUNOS: Juniper Operating System
- L-OSPF: Label OSPF
- LSP: Label Switch Path
- MAC-VRF, Media Access Control VRF
- MPLS: Multi-Protocol Label Switching
- OSPF: Open Shortest Path First
- P: Provider
- PE: Provider Edge
- RPL: Ring Protection Link
- RR: Route Reflector
- SID: Segment Identifier
- SR: Segment Routing
- SR-MPLS, Segment Routing - MPLS
- SRGB: Segment Routing Global Block
- VLAN, Virtual LAN
- VRF, Virtual Routing and Forwarding
- WTR: Wait-To-Restore timer (ERPS)



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